Global Stability of Deep Gaussian ReLU Networks

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Abstract

A bias-free ReLU network of width n and depth L with independent $\mathcal{N}(0, \alpha/n)$ weights has a global Lipschitz constant $K_{n,L}(\alpha)$, the worst stretching over all pairs of inputs for one draw of the weights. We prove, for every n, every $L \geq 2$ and every $\theta > 0$,

$$\mathbb{P}\big(K_{n,L}(\alpha) > 1\big) \leq 2\exp\big(n\,B_L(\alpha,\theta)\big), \qquad B_L = L\,H(1/L) + \log 5 + \theta \log 4 + L\,g_\alpha(\theta),$$

with H the binary entropy and g_α elementary, and for $\alpha > 2$ the converse bound $\mathbb{P}(K_{n,L}(\alpha) \leq 1) \leq 5\alpha^2 L/(n(\alpha-2)^2)$. The fixed-depth threshold α_L therefore tends to 2, with $\alpha_L \geq 2\exp[-(\sqrt{10} + o(1))\sqrt{\log L/L}]$. The proof turns on an exact identity for realizable activation patterns: averaging over the 2^{nL} sign gauges, which preserve the law of the weights, turns the realizability event into a deterministic orthant count divided by 2^{nL}. The same identity gives an exact finite matrix formula for mean strict-region counts. At fixed width, the zero-bias mean and global survival probability are asymptotic to positive constants times $(1 - 2^{-n})^L$; with independent centred nondegenerate Gaussian biases, the mean count instead tends to a finite limit.

The second result ties the layers: for one matrix W applied repeatedly, $x_{t+1} = \phi(Wx_t)$, and any fixed horizon T, $\max_{t \leq T} |2^t \|x_t\|^2 - 1| \leq \delta$ with probability at least $1 - Ce^{-cN}$ for every deterministic unit start. For a typical matrix, the bound holds outside a set of exponentially small measure under any prescribed input distribution independent of the matrix. Here the dependence of the trajectory on the matrix is handled by conditioning, made exact by a Gaussian innovation lemma for adaptive orthogonal queries. The two results share their object and their obstacle and resolve it in opposite ways; the paper explains why each method fits its question, and uses the second to sharpen the expansive side of the first to an exponential rate. Selected identities, bounds and norm statistics are checked numerically at small width.

Contents

1 The object, the question, and the answer

1.1 The object

Fix integers $n \geq 1$ and $L \geq 1$ and a number $\alpha > 0$. Let $W_1, \dots, W_L$ be independent $n \times n$ random matrices, every entry of every matrix independent with law $\mathcal{N}(0, \alpha/n)$. Write $\phi(t) = \max\{t, 0\}$ for the scalar ReLU and apply it to a vector coordinate by coordinate. Starting from an input $s \in \mathbb{R}^n$ set

$$x_0(s) = s, \qquad h_\ell(s) = W_\ell x_{\ell-1}(s), \qquad x_\ell(s) = \phi(h_\ell(s)), \qquad 1 \leq \ell \leq L, \tag{1}$$

and let $F^{(\alpha)}_{n,L}(s) = x_L(s)$. There are no biases, no training and no data: the network is L random matrices and the operation of clipping negative coordinates to zero. The factor $1/n$ in the variance is the usual width normalization; it keeps a coordinate of $W_\ell x$ at variance $(\alpha/n)\|x\|^2$, so that $\|W_\ell x\|^2$ has expectation $\alpha\|x\|^2$ rather than $n\alpha\|x\|^2$. The parameter α is the dial of the problem.

1.2 The question

A function $F : \mathbb{R}^n \to \mathbb{R}^n$ stretches or shrinks distances. Its global Lipschitz constant

$$K_{n,L}(\alpha) = \mathrm{Lip}_2\big(F^{(\alpha)}_{n,L}\big) = \sup_{s \neq t} \frac{\left\| F^{(\alpha)}_{n,L}(s) - F^{(\alpha)}_{n,L}(t) \right\|}{\|s - t\|} \tag{2}$$

is the worst stretching factor over every pair of inputs. It is not an average and not a typical value; one realization of the weights must control every pair at once. When $K_{n,L}(\alpha) \leq 1$ the network never increases a distance and we call it non-expansive. The question is: for which α is the random network non-expansive, and how does the answer depend on the width and the depth?

For a fixed depth L, define the fixed-depth threshold

$$\alpha_L = \sup\{\alpha > 0 : \ \mathbb{P}\big(K_{n,L}(\alpha) \le 1\big) \to 1 \text{ as } n \to \infty\}. \tag{3}$$

Membership in the set is itself a statement about a limit in the width, with α and L held fixed.

1.3 The answer, at finite width

The proof below produces, for every width and depth, an explicit bound on the probability that the network is expansive. Two elementary functions carry it. The binary entropy is

$$H(u) = -u \log u - (1-u)\log(1-u), \qquad 0 \le u \le 1, \qquad 0 \log 0 = 0,$$

and for $\theta > 0$ put

$$z_\theta = \frac{2}{1+\sqrt{1+8\theta}}, \qquad g_\alpha(\theta) = \theta \log \frac{\alpha}{e z_\theta^2 (1+z_\theta)} + \log \frac{1+z_\theta}{2z_\theta}. \tag{4}$$

The entropy will measure how many activation patterns a random network can realize; g_α will measure the high moments of one layer's radial factor.

Theorem 1.1 (finite width, finite depth)**.** *For every $n \ge 1$, every $L \ge 2$, every $\alpha > 0$ and every $\theta > 0$,*

$$\begin{gathered} \mathbb{P}\big(K_{n,L}(\alpha) > 1\big) \ \le \ \mathbb{E}\big[K_{n,L}(\alpha)^{2\theta n}\big] \ \le \ 2\exp\big(n\, B_L(\alpha,\theta)\big), \\ B_L(\alpha,\theta) = L\, H(1/L) + \log 5 + \theta \log 4 + L\, g_\alpha(\theta). \end{gathered} \tag{5}$$

For $L = 1$ the same holds with $LH(1/L)$ replaced by $\log 2$.

The four terms of the budget B_L have four separate origins, listed in section 6.2: the entropy term counts realizable activation patterns, $\log 5$ pays for a net of the sphere, $\theta \log 4$ for replacing an operator norm by a maximum over that net, and $L\, g_\alpha(\theta)$ bounds the radial moment exponent over L layers. The first three do not depend on α; the last is the only one that carries a factor L with a sign that can be negative. When $\alpha < 2$ the function g_α is negative on an interval $(0, \theta_0)$, because its derivative at zero is $\log(\alpha/2) < 0$, so the last term is a negative multiple of L while the positive terms grow like $\log L$. Beyond a computable depth the budget is negative.

Corollary 1.2 (fixed depth, growing width)**.** *Let $\alpha < 2$ and $L \ge 2$, and suppose $c_L(\alpha) := \sup_{\theta>0}\big[-B_L(\alpha,\theta)\big] > 0$. Then for every width n,*

$$\mathbb{P}\big(K_{n,L}(\alpha) > 1\big) \le 2e^{-c_L(\alpha)\, n},$$

so the network is non-expansive with probability at least $1 - 2e^{-c_L(\alpha)n}$. In particular $\mathbb{P}(K_{n,L}(\alpha) \le 1) \to 1$ as $n \to \infty$, with an exponential rate.

At a given finite width the probability is not one; the bound says how far from one it can be. For instance at $\alpha = 1$ and $L = 1000$ the certified rate is $c_{1000}(1) = 54.81$ to two decimal places, and already at width $n = 10$ the bound is less than $2e^{-548}$. Section 9 tabulates these numbers.

Theorem 1.3 (the other side, at finite width)**.** *For every $n \ge 1$, $L \ge 1$ and $\alpha > 2$,*

$$\mathbb{P}\big(K_{n,L}(\alpha) \le 1\big) \le \frac{5\alpha^2 L}{n(\alpha-2)^2}.$$

The two finite statements combine into the theorem the paper is named for.

Theorem 1.4 (unconditional global threshold)**.** *For the Gaussian bias-free ReLU network above:*

(i) For every $\alpha < 2$ there is a depth $L_0(\alpha)$ such that for every fixed $L \geq L_0$ there is $c_{\alpha,L} > 0$ with $\mathbb{P}(K_{n,L}(\alpha) > 1) \leq 2e^{-c_{\alpha,L}n}$ for all n; in particular $\mathbb{P}(K_{n,L}(\alpha) \leq 1) \to 1$.

(ii) For every fixed $L \geq 1$ and every $\alpha > 2$, $\mathbb{P}(K_{n,L}(\alpha) > 1) \to 1$ as $n \to \infty$.

(iii) Consequently $\lim_{L\to\infty} \alpha_L = 2$.

(iv) More precisely, $\alpha_L \geq 2\exp\left[-(\sqrt{10} + o(1))\sqrt{\log L/L}\right]$ as $L \to \infty$.

Part (i) is Corollary 1.2 together with the observation that the budget turns negative at large depth; the depth $L_0(\alpha)$ is the smallest L at which $c_L(\alpha) > 0$, and it is computable (section 9). Part (ii) is Theorem 1.3. Part (iii) follows from (i) and (ii). Part (iv) comes from expanding g_α near $\alpha = 2$, where $g_{2e^{-\delta}}(\theta) = -\delta\theta + \frac{5}{2}\theta^2 + O(\theta^3)$; optimizing θ and balancing the negative term $-L\delta^2/10$ against the entropy $\log L$ gives $\delta \approx \sqrt{10 \log L/L}$.

1.4 The order of limits

The order is part of the statement. First a depth L is fixed; then the width n tends to infinity, which defines α_L; only then does L tend to infinity. Nothing is claimed about a joint regime $L = L(n)$. Theorem 1.1 is nevertheless a finite statement in both variables, and one may read it in any regime one likes, provided the budget B_L is evaluated there. For example, at fixed $\alpha < 2$ the bound is useless below the depth $L_0(\alpha)$, whatever the width, which is a limitation of this certificate. There is also a genuine depth dependence: at depth one the threshold is $\alpha_1 = 1/4$, so a depth-one network at $\alpha = 1$ is expansive with probability tending to one although $1 < 2$.

1.5 Why the number two

For a fixed nonzero x, each coordinate of $W_\ell x$ is Gaussian with variance $(\alpha/n)\|x\|^2$, and ReLU keeps, in expectation, half of the squared magnitude:

$$\mathbb{E}\left[\|\phi(W_\ell x)\|^2 \;\middle|\; x\right] = \frac{\alpha}{2}\|x\|^2.$$

A forward pass multiplies a typical squared norm by $\alpha/2$ per layer, and $\alpha = 2$ is where that factor is one. The content of the theorem is that the same number governs the global Lipschitz constant in the deep limit, although the supremum in (2) may hunt for rare activation regions and rare directions, and although the region an input lands in is decided by the very matrices whose stretching is being measured.

1.6 The second result: one matrix, many steps

The network above draws a fresh matrix at every layer. Tie the layers instead, one matrix W with $\mathcal{N}(0, 1/N)$ entries applied again and again, $x_{t+1} = \phi(Wx_t)$ from a unit vector x_0; in the notation above this is $\alpha = 1$ and $n = N$. At $\alpha = 1$ an independent Gaussian layer halves the conditional expected squared norm. For the tied iteration, the second theorem (section 7) gives the same norm profile to any tolerance, uniformly over a fixed number of steps, with exponentially small failure probability: for every T and δ there are C, c depending only on them with

$$\mathbb{P}\Big(\max_{0\leq t\leq T}\left|2^t\|x_t\|^2 - 1\right| > \delta\Big) \leq Ce^{-cN}$$

for every deterministic x_0; and for one typical matrix the set of starting points that fail has exponentially small measure under any measure independent of W. The obstacle is the same as before: x_1 is a function of W, so Wx_1 is not a fresh Gaussian vector. The resolution is the opposite of the gauge argument. The trajectory spans a subspace of dimension at most $T+1$; a query Wu reveals only the coefficients of the rows of W in the direction u; and if the directions queried are orthogonal and each is chosen from what has already been revealed, the answers are independent Gaussian vectors, whatever the rule that chose them. Gram–Schmidt on the trajectory produces such directions, the dynamics is rewritten in terms of $T+1$ Gaussian innovations, and a law of large numbers uniform over a ball in $\mathbb{R}^{T+1}$ closes the argument. Section 8 sets the two results side by side: what they share, where they differ, why each method fits its question, and one place where the second method improves the first theorem.

1.7 What is new here

The first theorem's proof turns on the exact orbit identity of section 4: the probability that a formal activation pattern is realized, weighted by any gauge-invariant functional, equals the expected number of orthants met by a random subspace divided by 2^{nL}. Nothing is conditioned on and no independence is assumed; the identity is exact and the only inequality is a deterministic orthant count. Written at finite width, the argument gives an explicit failure bound for every n and L, with a budget whose four terms have four named origins, and a computable sufficient finite-depth threshold; the limit theorem and a lower bound on the threshold are read off from it. The second theorem's proof turns on the innovation lemma of section 7, which makes conditioning on an adaptively chosen trajectory exact. Placing the two together shows that the same entanglement admits two different exact treatments, that the choice between them is decided by what is quantified over, one input or all inputs, and that the second treatment sharpens the expansive side of the first theorem.

At fixed width, the same identity gives an exact finite matrix formula for the mean strict-region count, with separate formulas for zero biases and independent centred nondegenerate Gaussian biases (Theorem 10.4). In the zero-bias case, both global survival probability and the mean strict count, divided by $(1-2^{-n})^L$, increase to finite positive limits; the mean constant is an explicit rational number (Corollary 10.6). With nondegenerate biases, the expected count instead increases to a finite rational limit (Corollary 10.7). These are statements about strict activation regions and a different order of limits from the threshold theorem.

1.8 Related work and the scope of the contribution

Global Lipschitz bounds for random ReLU networks already have a substantial literature. Geuchen, Stöger, Telaar and Voigtlaender [1] study scalar-output maps $\Phi : \mathbb{R}^d \to \mathbb{R}$ with hidden width N, Gaussian weights and independent biases. Their Theorems 3.3 and 3.4 give upper and lower bounds for deep networks: the upper bound assumes $N > d+2$, and the lower bound assumes $N \gtrsim dL^2$. Their hidden-weight variance is $2/N$ and their final scalar readout has variance one; normalization therefore matters when comparing numerical Lipschitz constants. Our map has input, hidden and output dimension all equal to n, variance α/n, and ReLU at the output. We control its Euclidean operator norm globally and obtain a sufficient contraction certificate at every finite width. The cited deep-network theorems cannot simply be specialized to this square regime by setting $d = N$.

Dirksen, Finke, Geuchen, Stöger and Voigtlaender [2] sharpen the scalar-output theory for all ℓ^p input metrics. In the zero-bias case, their Corollary 4.2 gives, for $p = 2$, an upper bound of order $\sqrt{d \log(N/d)}$ under a width condition of order $N \gtrsim dL^3$, up to logarithmic factors; their Corollary 3.6 gives a lower bound of order $\sqrt{d}$ under its hypotheses. Their global upper bound combines gradient estimates with control of nearby activation changes using random hyperplane tessellations. Our argument instead averages strict realizability over sign gauges, retaining its dependence on the Jacobian functional, before applying a net to a fixed formal linear map. The different dimensions, readout and width hypotheses prevent a direct numerical comparison or a claim that one theorem improves the other.

For tied weights, Yang's Tensor Programs I [3], in particular Theorem 5.4 and its recurrent-network examples, supplies a broad framework for Gaussian limits and convergence of empirical coordinate averages in fixed programs. Reusing Gaussian matrices and deriving deterministic norm recursions are therefore not new in themselves; Appendix G.3 also presents the Gaussian conditioning method used in that theory. Our finite-horizon statement gives a quantitative failure bound Ce^{-cN} for this particular ReLU iteration, uniformly over a fixed time horizon and uniformly in the choice of deterministic unit start. A Fubini argument then bounds the measure of bad starts for a typical matrix. Neither this conclusion nor the cited fixed-program limit supplies uniform control over every input of one matrix or all-time convergence.

Region counts at initialization are an established topic. Hanin and Rolnick [4] study counts along one-dimensional subspaces and the volume of region boundaries. Their later paper [5], Theorem 5, bounds expected activation-region counts in input cubes under gradient and bias-density assumptions; Appendix D compares counts before and after removing biases. Wang [8], Theorem 2.1, bounds expected maximal linear-region counts for several architectures, including width two, with independent continuously distributed symmetric weights and biases and scalar output. Kogan, Jananthan and Kepner [6], Theorem IV.2, study expected output breakpoints for scalar input and output as hidden widths grow. The exact formulas here count strict activation regions of the full square input space at finite width and depth, keeping different activation patterns even when their final output maps agree.

Collapse at initialization is also established. Lu, Shin, Su and Karniadakis [9] analyze the probability that a randomly initialized ReLU network is constant under symmetric initialization. Rister and Rubin [10] give geometric whole-network survival bounds and a relative-convergence argument for finite data in a growing-width regime. Lu et al., Theorem 3.5 and Appendix C, give one-dimensional-input bounds and a one-ray state analysis. Corollary 10.6 uses the finite transfer formula to control the normalized strict-region mean and survival probability on the whole square input space; the discussion there compares the earlier ratio arguments explicitly.

The contribution claimed here is the weighted sign-gauge identity and the finite-width budget it yields for square vector-output networks, together with a self-contained quantitative finite-horizon argument and an explicit comparison of their quantifiers. The rate in Theorem 1.4(iv) is a certified lower bound on α_L, not a matching asymptotic rate for the true threshold.

1.9 How the proof goes

The difficulty is an entanglement. The activation pattern of an input is a function of the weights; the worst input is chosen after the weights are seen; and the Jacobian on the region of that input is built from the same weights. Conditioning on a realized pattern would tilt the law of the weights in a way that is hard to control. The proof never conditions. Its steps:

1. Replace the global supremum by the maximum operator norm of the linear maps on the nonempty activation regions (section 3).

2. Fix a *formal* mask sequence D: a sequence of diagonal 0/1 matrices written down without asking whether any input produces it. Realizability of D is the event that an n-dimensional subspace of $\mathbb{R}^{nL}$ meets one specified open orthant.

3. Average that event over a group of 2^{nL} sign changes. The sign changes preserve the joint law of the weights and the norm of every formal Jacobian, while they move the target orthant through all 2^{nL} orthants exactly once. The average of the realizability indicator

is therefore a deterministic orthant count divided by 2^{nL}, and an exact identity results (section 4). This identity is the centre of the argument.

4. Bound the orthant count by a classical result: an at most n-dimensional subspace of $\mathbb{R}^{nL}$ meets at most $C_{n,L} = 2\sum_{j<n}\binom{nL-1}{j} \le 2e^{nLH(1/L)}$ open orthants.
5. Bound the operator norm of a formal Jacobian through a $\frac{1}{2}$-net of the sphere and exact Gaussian radial moments, and reduce everything to one scalar random variable $X_n^{(\alpha)} = \frac{\alpha}{n}\sum_i B_i Z_i^2$ (section 5).
6. Take a moment of order $2\theta n$ and bound it with an explicit finite-n inequality; the four exponential rates assemble into B_L (section 6).

Every estimate in these six steps is finite in n and L. The threshold α_L and asymptotic form (iv) are then read from the finite bound; the latter uses an expansion in θ. Section 9 checks the identity of step 3 and the bounds of steps 4–6 against brute-force computation at small width. The scalar function g_α does two jobs: it bounds the moment at every finite width, and section 10 shows that it is also its exact limiting exponent. Thus no asymptotic approximation is needed in the finite theorem.

2 Tools

Everything in this section is standard. It is collected so that the later sections can refer to one place and so that the paper reads without outside references.

2.1 Norms

For $x \in \mathbb{R}^n$, $\|x\| = \left(\sum_i x_i^2\right)^{1/2}$ and $\langle x, y\rangle = x^T y$; the Cauchy–Schwarz inequality $|\langle x, y\rangle| \le \|x\|\,\|y\|$ says a cosine is at most one. The unit sphere is $S^{n-1} = \{x : \|x\| = 1\}$. For a linear map $T : \mathbb{R}^n \to \mathbb{R}^m$ the operator norm is

$$\|T\|_{2\to 2} = \sup_{x\neq 0}\frac{\|Tx\|}{\|x\|} = \sup_{\|x\|=1}\|Tx\|,$$

the two forms agreeing because $x = \|x\|\cdot x/\|x\|$. Thus $\|Tx\| \le \|T\|_{2\to 2}\|x\|$ for every x. The supremum is attained on the compact sphere; a unit vector v with $\|Tv\| = \|T\|_{2\to 2}$ is a right singular vector for the largest singular value. If Q is orthogonal, $Q^TQ = I$, then $\|Qx\|^2 = x^TQ^TQx = \|x\|^2$, so $\|QT\|_{2\to 2} = \|T\|_{2\to 2} = \|TQ\|_{2\to 2}$. A diagonal matrix with entries ± 1 is orthogonal; this is the one fact about orthogonal matrices the proof uses.

2.2 ReLU

$\phi(t) = \max\{t, 0\}$, applied coordinatewise to vectors.

Lemma 2.1. *For all real a, b and every $c \ge 0$: $|\phi(a) - \phi(b)| \le |a - b|$, $\phi(ca) = c\phi(a)$, and $\phi(a) = \mathbf{1}_{\{a>0\}}a$ whenever $a \neq 0$. Consequently $\|\phi(x) - \phi(y)\| \le \|x - y\|$ for vectors.*

Proof. If $a, b \ge 0$ the first inequality is an equality; if $a, b \le 0$ both values are zero; if $a \ge 0 \ge b$ then $|\phi(a) - \phi(b)| = a \le a - b$. Homogeneity holds because multiplying by $c \ge 0$ does not change the sign of a, and the indicator formula is the definition away from zero. For vectors, square the scalar inequality coordinatewise and sum. □

Homogeneity is what makes a bias-free network positively homogeneous: from (1), $x_\ell(cs) = cx_\ell(s)$ for $c \ge 0$ by induction on ℓ, so

$$F_{n,L}^{(\alpha)}(cs) = cF_{n,L}^{(\alpha)}(s), \qquad c \ge 0, \qquad \text{and} \qquad F_{n,L}^{(\alpha)}(0) = 0. \tag{6}$$

2.3 Lipschitz constants and Jacobians

$F : \mathbb{R}^n \to \mathbb{R}^m$ is globally Lipschitz if $\|F(s) - F(t)\| \le K \|s - t\|$ for all s, t; the least such K is $\mathrm{Lip}_2(F)$. If F is differentiable at s with Jacobian $DF(s)$, then for a unit vector v,

$$\|DF(s)v\| = \lim_{h\to 0} \frac{\|F(s + hv) - F(s)\|}{|h|} \le \mathrm{Lip}_2(F),$$

so $\|DF(s)\|_{2\to 2} \le \mathrm{Lip}_2(F)$ at every point of differentiability. For a continuous map that is linear on each of finitely many polyhedral pieces the reverse also holds, and equality is exactly what the network needs; that is Lemma 3.4 below.

The constant $K_{n,L}(\alpha)$ of (2) is finite: ReLU is 1-Lipschitz, so

$$\|x_\ell(s) - x_\ell(t)\| \le \|W_\ell\|_{2\to 2} \|x_{\ell-1}(s) - x_{\ell-1}(t)\| ,$$

and iterating gives $K_{n,L}(\alpha) \le \prod_\ell \|W_\ell\|_{2\to 2} < \infty$. It is measurable: by continuity the supremum in (2) may be restricted to rational pairs, and each ratio is a Borel function of the weights.

2.4 Three probability inequalities

Lemma 2.2. *(a) Union bound:* $\mathbb{P}(\bigcup_j A_j) \le \sum_j \mathbb{P}(A_j)$. *(b) Markov: if* $Y \ge 0$ *and* $a > 0$ *then* $\mathbb{P}(Y > a) \le \mathbb{E}Y/a$*; in particular* $\mathbb{P}(Y > 1) \le \mathbb{E}Y$. *(c) Chebyshev: if* Y *has finite variance then* $\mathbb{P}(|Y - \mathbb{E}Y| > \varepsilon) \le \mathrm{Var}(Y)/\varepsilon^2$.

Proof. (a) $\mathbf{1}_{\cup A_j} \le \sum \mathbf{1}_{A_j}$ pointwise; take expectations. (b) $Y \ge a\mathbf{1}_{\{Y>a\}}$ pointwise; take expectations and divide by a. (c) Apply (b) to $(Y - \mathbb{E}Y)^2$ with threshold ε^2. $\square$

2.5 Gaussian facts

$Z \sim \mathcal{N}(0, 1)$ has density $(2\pi)^{-1/2}e^{-z^2/2}$, $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 = 3$ (two integrations by parts), and for $t < 1/2$

$$\mathbb{E}e^{tZ^2} = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(1-2t)z^2/2}\,\mathrm{d}z = (1 - 2t)^{-1/2}. \tag{7}$$

If $G \in \mathbb{R}^m$ has independent $\mathcal{N}(0, 1)$ coordinates, $\|G\|^2$ has the chi-square law with m degrees of freedom, written χ^2_m; we set $\chi^2_0 = 0$. If Q is a fixed orthogonal matrix then QG is again centred Gaussian with covariance $QQ^T = I$, so $QG \overset{\mathrm{d}}{=} G$; in particular multiplying rows or columns of a Gaussian matrix by fixed signs does not change its law. A centred Gaussian vector is determined by its covariance, so jointly Gaussian variables with zero covariance are independent.

2.6 A bound on partial binomial sums

Lemma 2.3. *For integers* $N \ge 1$ *and* $0 \le k \le N/2$*,*

$$\sum_{j=0}^{k} \binom{N}{j} \le \exp\big(N\,H(k/N)\big).$$

Proof. For $k = 0$ both sides are 1. Otherwise put $q = k/N \in (0, 1/2]$. Expanding $1 = (q + (1 - q))^N$ and keeping only the terms $j \le k$,

$$1 \ge \sum_{j\le k} \binom{N}{j} q^j (1 - q)^{N-j} \ge \sum_{j\le k} \binom{N}{j} q^k (1 - q)^{N-k},$$

because $q \le 1-q$ makes $q^j(1-q)^{N-j}$ non-increasing in j. Hence $\sum_{j\le k} \binom{N}{j} \le q^{-k}(1-q)^{-(N-k)} = \exp(NH(q))$. $\square$

This is the only counting estimate the finite theorem needs. Stirling's formula appears in section 10, where the exact exponent is computed, and nowhere else.

2.7 One deterministic inequality

Lemma 2.4. *For every $x \ge 0$, $p > 0$ and $t > 0$, $x^p \le (p/(et))^p e^{tx}$.*

Proof. For $x > 0$, $\log(x^p e^{-tx}) = p \log x - tx$ has derivative $p/x - t$, positive before $x = p/t$ and negative after, so the maximum of $x^p e^{-tx}$ is $(p/t)^p e^{-p}$. Multiply by e^{tx}. □

The inequality is tight at exactly one point, $x = p/t$; it is what converts a moment of order p into a moment-generating function, at the price of the factor $(p/(et))^p$.

3 Regions: from a global supremum to formal masks

3.1 Activation masks and the Jacobian

At an input s for which no coordinate of any preactivation $h_\ell(s)$ is zero, define the activation masks

$$D_\ell(s) = \operatorname{diag}\big(\mathbf{1}_{\{h_{\ell 1}(s)>0\}}, \dots, \mathbf{1}_{\{h_{\ell n}(s)>0\}}\big), \qquad 1 \le \ell \le L,$$

diagonal 0/1 matrices recording which coordinates the ReLU keeps. Away from zero the derivative of the scalar ReLU is 1 on the positive half-line and 0 on the negative one, so the chain rule gives

$$J(s) = D_L(s)\, W_L\, D_{L-1}(s)\, W_{L-1} \cdots D_1(s)\, W_1. \tag{8}$$

The network is linear in a neighbourhood of such an s, with matrix $J(s)$. The difficulty named in the introduction is visible in (8): the masks are functions of the weights, and the input that produces the worst mask is chosen after the weights are drawn.

3.2 Formal masks

The step that breaks the entanglement is to stop insisting that a mask come from an input.

Definition 3.1 (formal mask sequence)**.** A formal mask sequence is a tuple $D = (D_1, \dots, D_L)$ of $n \times n$ diagonal matrices with diagonal entries in $\{0, 1\}$. There are 2^n choices at each layer and 2^{nL} sequences in all.

The word formal means that D is chosen on paper; no input is claimed to produce it. For a fixed D define the formal preactivation matrices by

$$H_1^D = W_1, \qquad H_\ell^D = W_\ell D_{\ell-1} H_{\ell-1}^D \quad (2 \le \ell \le L), \qquad \text{so} \qquad H_\ell^D = W_\ell D_{\ell-1} W_{\ell-1} \cdots D_1 W_1, \tag{9}$$

and the formal Jacobian

$$J_D = D_L H_L^D = D_L W_L D_{L-1} W_{L-1} \cdots D_1 W_1. \tag{10}$$

Compare with (8): the shape is identical, and the whole difference lies in where the masks came from.

To say "the signs of a vector y agree with the mask D_ℓ" with one inequality, put $E_\ell^D = 2D_\ell - I_n$, a diagonal matrix with entries $+1$ where the mask is 1 and -1 where it is 0. Then

$$E_\ell^D\, y > 0 \text{ coordinatewise} \quad \Longleftrightarrow \quad y_i > 0 \text{ where } (D_\ell)_{ii} = 1 \ \text{ and } \ y_i < 0 \text{ where } (D_\ell)_{ii} = 0.$$

Definition 3.2 (strict region)**.** The strict region of a formal mask sequence D is

$$R_D = \big\{ s \in \mathbb{R}^n : \ E_\ell^D H_\ell^D s > 0 \text{ coordinatewise for every } 1 \le \ell \le L \big\}. \tag{11}$$

Every condition in (11) is a strict homogeneous linear inequality in s, so R_D is an open convex cone, possibly empty. It is open because strict inequalities survive small perturbations, convex because the inequalities are linear, and a cone because they are homogeneous.

Lemma 3.3 (consistency). *If $s \in R_D$ then the forward pass at s has mask D_ℓ at every layer, and*

$$h_\ell(s) = H_\ell^D s, \qquad x_\ell(s) = D_\ell H_\ell^D s, \qquad 1 \le \ell \le L.$$

In particular $F_{n,L}^{(\alpha)}(s) = J_D s$ and $DF_{n,L}^{(\alpha)}(s) = J_D$. Conversely, an input at which every preactivation coordinate is nonzero lies in exactly one strict region.

Proof. Induction on the layer. At layer one, $h_1(s) = W_1 s = H_1^D s$ by definition. Since $s \in R_D$, $E_1^D H_1^D s > 0$ coordinatewise: where $(D_1)_{ii} = 1$ this says $(H_1^D s)_i > 0$ and ReLU keeps the coordinate; where $(D_1)_{ii} = 0$ it says $(H_1^D s)_i < 0$ and ReLU sends it to zero. Coordinate by coordinate, $x_1(s) = \phi(h_1(s)) = D_1 H_1^D s$. Suppose the claim through layer $\ell - 1$. Then $h_\ell(s) = W_\ell x_{\ell-1}(s) = W_\ell D_{\ell-1} H_{\ell-1}^D s = H_\ell^D s$ by (9), and the inequality $E_\ell^D H_\ell^D s > 0$ says the signs of $h_\ell(s)$ are those encoded by D_ℓ, so $x_\ell(s) = D_\ell H_\ell^D s$. At the last layer $F(s) = x_L(s) = D_L H_L^D s = J_D s$. The strict inequalities hold in a neighbourhood of s, so the network is the linear map J_D there and its derivative at s is J_D.

For the converse, take an input with all preactivation coordinates nonzero and define $(D_\ell)_{ii} = \mathbf{1}_{\{h_{\ell i}(s) > 0\}}$. The induction just made shows $h_\ell(s) = H_\ell^D s$ and every inequality of (11) holds, so $s \in R_D$; and the sign of each nonzero $h_{\ell i}(s)$ determines $(D_\ell)_{ii}$, so D is unique. $\square$

3.3 Realizability as a subspace meeting an orthant

Stack the formal preactivations:

$$A_D = \begin{pmatrix} H_1^D \\ H_2^D \\ \vdots \\ H_L^D \end{pmatrix} \in \mathbb{R}^{nL \times n}, \tag{12}$$

so that $A_D s$ lists all L formal preactivation vectors of s on top of one another. Let

$$O_D = \{(y_1, \dots, y_L) \in (\mathbb{R}^n)^L : \ E_\ell^D y_\ell > 0 \text{ for every } \ell\},$$

one open coordinate orthant of $\mathbb{R}^{nL}$. Then, directly from the definitions,

$$R_D \neq \varnothing \quad \Longleftrightarrow \quad \operatorname{im}(A_D) \cap O_D \neq \varnothing. \tag{13}$$

This is the geometric form of realizability that the symmetry argument of section 4 works with: the mask D is realized by some input exactly when the image of A_D, a subspace of dimension at most n in a space of dimension nL, touches one specified orthant.

The event $\{R_D \neq \varnothing\}$ is measurable. Write $M = nL$, let $\tau_D \in \{\pm 1\}^M$ be the sign vector of O_D, and for $A \in \mathbb{R}^{M \times n}$ put $\Psi_D(A) = \max_{\|s\|=1} \min_i (\tau_D)_i (As)_i$, a maximum of a continuous function over the compact sphere. If $As \in O_D$ for some s then $s \neq 0$ and, by homogeneity, $\Psi_D(A) > 0$; conversely a maximizing unit vector shows $\Psi_D(A) > 0$ implies $\operatorname{im} A \cap O_D \neq \varnothing$. And $|\Psi_D(A) - \Psi_D(B)| \le \sup_{\|s\|=1} \|(A - B)s\|_\infty$, so Ψ_D is continuous and $\{\Psi_D > 0\}$ is open. Since A_D is a measurable function of the weights, so is the indicator of realizability.

3.4 The Lipschitz constant of a piecewise-linear map

Lemma 3.4. *Let $F : \mathbb{R}^n \to \mathbb{R}^m$ be continuous, and suppose a finite polyhedral complex covers $\mathbb{R}^n$ such that on every full-dimensional cell C the map F agrees with a linear map M_C. Then*

$$\operatorname{Lip}_2(F) = \max_C \|M_C\|_{2 \to 2}.$$

Why this is not obvious: the Lipschitz constant compares two points that may be far apart, with many cells between them, while the operator norms are local statements about single cells. Nothing in the definition says a journey across a hundred cells cannot accumulate more stretching than any one cell allows. The proof shows it cannot, and the reason is that the pieces of a straight segment add up to exactly the length of the segment.

Proof. Let $M = \max_C \|M_C\|_{2\to 2}$. Fix s, t and parameterize the segment $\gamma(u) = s + u(t-s)$, $0 \le u \le 1$. A finite complex has finitely many faces, each cut out by finitely many linear equations and inequalities. Thus the parameter interval has a finite subdivision on whose open subintervals the minimal face containing the segment is constant: $[0,1]$ splits as $0 = u_0 < u_1 < \cdots < u_r = 1$ with the interior of each $\gamma([u_{j-1}, u_j])$ inside one face. Choose a full-dimensional cell C_j whose closure contains that face. On the interior of C_j, $F = M_{C_j}$; both sides are continuous, so they agree on the closure, in particular at the two endpoints of the subsegment. Therefore

$$
\begin{aligned}
\|F(t) - F(s)\| &\le \sum_{j=1}^{r} \|F(\gamma(u_j)) - F(\gamma(u_{j-1}))\| = \sum_{j=1}^{r} \left\| M_{C_j}\big(\gamma(u_j) - \gamma(u_{j-1})\big) \right\| \\
&\le \sum_{j=1}^{r} \left\|M_{C_j}\right\|_{2\to 2} \|\gamma(u_j) - \gamma(u_{j-1})\| \le M \sum_{j=1}^{r} (u_j - u_{j-1}) \|t - s\| = M \|t - s\| .
\end{aligned}
$$

The last equality is the point: along a straight line with constant velocity the displacements are $(u_j - u_{j-1})(t-s)$, and the parameter increments sum to one, so the pieces cannot accumulate. Hence $\mathrm{Lip}_2(F) \le M$.

For the reverse inequality choose a cell C with $\|M_C\|_{2\to 2} = M$, an interior point $s \in C$, and a unit v with $\|M_C v\| = M$. For small h, $s + hv \in C$, so $\|F(s+hv) - F(s)\| / |h| = \|M_C v\| = M$, and $\mathrm{Lip}_2(F) \ge M$. $\square$

3.5 Why the network has finitely many pieces, and the nondegeneracy event

Each coordinate of $W_1 s$ changes sign only across a hyperplane; on a region of fixed signs the first layer is multiplication by a fixed diagonal 0/1 matrix, hence linear. Repeating the argument layer by layer refines $\mathbb{R}^n$ into finitely many polyhedral cells on which all signs are fixed and the network is linear; there are at most 2^{nL} sign sequences, so finitely many pieces.

One technical point: a preactivation functional could vanish identically on a cell, in which case the interior of that cell is not a strict region. With probability one this happens only after the formal trajectory has already died. Fix D, set $B_0^D = I_n$ and $B_{\ell-1}^D = D_{\ell-1} H_{\ell-1}^D$ for $\ell \ge 2$, so $H_\ell^D = W_\ell B_{\ell-1}^D$. Condition on $W_1, \ldots, W_{\ell-1}$; then $B_{\ell-1}^D$ is fixed and row i of H_ℓ^D is $w_i^T B_{\ell-1}^D$ with w_i the i-th row of W_ℓ. If $B_{\ell-1}^D \ne 0$, the map $w \mapsto w^T B_{\ell-1}^D$ is not zero, its kernel is a proper subspace, and a Gaussian vector lies in a proper subspace with probability zero. There are finitely many (D, ℓ, i), so almost surely: whenever $B_{\ell-1}^D \ne 0$, every row of H_ℓ^D is a nonzero functional. If instead $B_{\ell-1}^D = 0$ then $H_\ell^D = 0$ and everything after it, including J_D, is zero: a dead formal piece.

Lemma 3.5 (exact reduction)**.** *With probability one over the weights,*

$$
K_{n,L}(\alpha) = \max\big(\{0\} \cup \{\|J_D\|_{2\to 2} : R_D \ne \varnothing\}\big). \tag{14}
$$

Consequently, for every $t \ge 0$,

$$
\{K_{n,L}(\alpha) > t\} \subseteq \bigcup_D \{R_D \ne \varnothing,\ \|J_D\|_{2\to 2} > t\}. \tag{15}
$$

Proof. Work on the nondegeneracy event. Refine input space into the finite complex on whose full-dimensional cells the network is linear. Take one such cell. If its formal forward map has not died, every preactivation functional used to define the next sign split is nonzero, its zero set is a hyperplane with empty interior, and the interior of each resulting full-dimensional subcell has strict signs at every layer; by Lemma 3.3 that subcell lies in a nonempty R_D and the network's linear map there is J_D. If the formal map dies, bias-freeness makes every later output zero on the cell, so its linear map is the zero map. Thus every full-dimensional linear piece has matrix J_D for a nonempty strict region or 0, and Lemma 3.4 gives (14). If $K_{n,L}(\alpha) > t \geq 0$, the maximum exceeds t, the zero cannot be responsible, so some D has $R_D \neq \varnothing$ and $\|J_D\|_{2\to 2} > t$. □

The lemma turns a supremum over all pairs of inputs into a maximum over finitely many objects, each attached to a formal mask. Boundary points between cells do not matter: the segment proof of Lemma 3.4 is the reason.

4 The symmetry, the realizable masks, and the counting

After Lemma 3.5 the problem is a maximum over the 2^{nL} formal masks, each contributing only if it is realized. Two things are still entangled: whether D is realized, and how large $\|J_D\|_{2\to 2}$ is, are both functions of the same weights. This section separates them by an exact identity, not by an independence assumption.

4.1 The gauge group

For each layer let Σ_ℓ be a diagonal matrix with diagonal entries in $\{-1, +1\}$, and set $\Sigma_0 = I_n$. Define transformed weights

$$W_1^\Sigma = \Sigma_1 W_1, \qquad W_\ell^\Sigma = \Sigma_\ell W_\ell \Sigma_{\ell-1} \quad (2 \leq \ell \leq L). \tag{16}$$

Because $\Sigma_{\ell-1}^{-1} = \Sigma_{\ell-1}$, the same formula reads as a change of sign of the hidden coordinates at layer $\ell - 1$: the sign inserted on the left of one layer is cancelled on the right of the next. The tuples $\Sigma = (\Sigma_1, \dots, \Sigma_L)$ form a group $\mathcal{G}$ with $|\mathcal{G}| = 2^{nL}$ elements. Write $S_\Sigma = \operatorname{diag}(\Sigma_1, \dots, \Sigma_L) \in \mathbb{R}^{nL\times nL}$.

Example 4.1. With $L = 2$, $W_1^\Sigma = \Sigma_1 W_1$ and $W_2^\Sigma = \Sigma_2 W_2 \Sigma_1$. The factor Σ_1 flips the signs of the first-layer preactivations; the matching factor on the right of W_2 cancels it inside the product; only Σ_2 survives in the formal Jacobian.

Lemma 4.2 (gauge covariance)**.** *Fix a formal mask sequence D. For every $\Sigma \in \mathcal{G}$,*

$$H_\ell^D(W^\Sigma) = \Sigma_\ell H_\ell^D(W), \qquad A_D(W^\Sigma) = S_\Sigma A_D(W), \qquad J_D(W^\Sigma) = \Sigma_L J_D(W). \tag{17}$$

Moreover $(W_1^\Sigma, \dots, W_L^\Sigma) \overset{\mathrm{d}}{=} (W_1, \dots, W_L)$.

Proof. At layer one, $H_1^D(W^\Sigma) = \Sigma_1 W_1 = \Sigma_1 H_1^D(W)$. If the first identity holds at layer $\ell - 1$, then

$$\begin{aligned} H_\ell^D(W^\Sigma) &= W_\ell^\Sigma D_{\ell-1} H_{\ell-1}^D(W^\Sigma) = \Sigma_\ell W_\ell \Sigma_{\ell-1} D_{\ell-1} \Sigma_{\ell-1} H_{\ell-1}^D(W) \\ &= \Sigma_\ell W_\ell D_{\ell-1} H_{\ell-1}^D(W) = \Sigma_\ell H_\ell^D(W), \end{aligned}$$

because diagonal matrices commute and $\Sigma_{\ell-1}^2 = I$. Stacking gives the second identity; the third follows since D_L and Σ_L commute. For the law: each entry of W_ℓ^Σ is the corresponding entry of W_ℓ times a fixed sign, a centred Gaussian has the same law after a sign change, and the transformation is a bijective coordinatewise sign change, so independence within and between matrices is preserved. □

A nonnegative measurable function $F(W_1,\dots,W_L)$ is gauge invariant if $F(W^\Sigma)=F(W)$ for all Σ. Since Σ_L is orthogonal, $\left\|J_D(W^\Sigma)\right\|_{2\to2}=\|\Sigma_L J_D(W)\|_{2\to2}=\|J_D(W)\|_{2\to2}$, and likewise $\left\|J_D(W^\Sigma)v\right\|=\|J_D(W)v\|$ for a fixed vector v: every nonnegative function of $\|J_D\|_{2\to2}$ or of $\|J_Dv\|$ is gauge invariant. The gauge changes the coordinates in which hidden representations are written; the internal signs cancel from the matrix product, and at the output only an orthogonal sign matrix remains, which cannot change a Euclidean norm.

4.2 How many orthants can a subspace meet

An open coordinate orthant of $\mathbb{R}^M$ is $O_\tau=\{x:\tau_i x_i>0\ \forall i\}$ for a sign vector $\tau\in\{\pm1\}^M$; there are 2^M of them. For a linear subspace $U\subseteq\mathbb{R}^M$ let

$$r(U)=\#\{\text{open orthants that meet } U\}.$$

If U lies in a coordinate hyperplane then $r(U)=0$. The fact that drives the proof is that a d-dimensional subspace cannot pass through all 2^M orthants when $d\ll M$; it has only d degrees of freedom and cannot choose all M signs independently.

Assume no coordinate functional vanishes identically on U. The M hyperplanes $\{x_i=0\}$ restrict to central hyperplanes in U and cut U into connected open cones. Two points of the same component have the same signs, since a sign change requires passing through zero; conversely the points of U with a fixed strict sign pattern form the convex, hence connected, set $U\cap O_\tau$. So $r(U)$ is the number of regions into which the coordinate hyperplanes cut U, and the question becomes the classical count of regions of a central arrangement.

Let $R(M,d)$ be the largest number of regions that M central hyperplanes can form in a space of dimension at most d. Adding the M-th hyperplane to an arrangement of $M-1$: a region it does not touch stays one region; a region it cuts becomes two, and the cut regions correspond one to one to the regions of the arrangement induced on the new hyperplane, which has dimension at most $d-1$ and carries at most $M-1$ central hyperplanes. Therefore

$$R(M,d)\le R(M-1,d)+R(M-1,d-1),\qquad R(M,1)\le 2,\qquad R(1,d)\le 2. \tag{18}$$

Define $\widetilde{R}(M,d)=2\sum_{j=0}^{d-1}\binom{M-1}{j}$, binomials outside their range being zero. Pascal's identity $\binom{M-1}{j}=\binom{M-2}{j}+\binom{M-2}{j-1}$ gives

$$\widetilde{R}(M-1,d)+\widetilde{R}(M-1,d-1)=2\sum_{j=0}^{d-1}\Big[\binom{M-2}{j}+\binom{M-2}{j-1}\Big]=2\sum_{j=0}^{d-1}\binom{M-1}{j}=\widetilde{R}(M,d),$$

and $\widetilde{R}(M,1)=\widetilde{R}(1,d)=2$, so induction on (18) gives $R(M,d)\le\widetilde{R}(M,d)$.

Lemma 4.3 (orthant count)**.** *Let $U\subseteq\mathbb{R}^M$ be a linear subspace of dimension $d\le n$. Then $r(U)\le 2\sum_{j=0}^{d-1}\binom{M-1}{j}\le 2\sum_{j=0}^{n-1}\binom{M-1}{j}$.*

Proof. If U is inside a coordinate hyperplane, $r(U)=0$. Otherwise the region count above gives the first inequality, and $d\le n$ gives the second. □

For the network $M=nL$ and $U=\mathrm{im}(A_D)$ has dimension at most n. Define

$$C_{n,L}=2\sum_{j=0}^{n-1}\binom{nL-1}{j},\qquad q_{n,L}=\frac{C_{n,L}}{2^{nL}}, \tag{19}$$

so that $r(\mathrm{im}\,A_D)\le C_{n,L}$ always. Two values are exact: $C_{n,1}=2^n$ (every orthant of $\mathbb{R}^n$ is available to a full-rank W_1) and $C_{n,2}=2^{2n-1}$ (half of 2^{2n-1} binomials of an odd row).

Lemma 4.4 (entropy bound)**.** *For $L \geq 2$, $C_{n,L} \leq 2\exp\big(nL\,H(1/L)\big)$.*

Proof. Apply Lemma 2.3 with $N = nL - 1$ and $k = n - 1$: since $k/N \leq 1/L \leq 1/2$ and H is increasing on $[0, 1/2]$, $\sum_{j \leq n-1} \binom{nL-1}{j} \leq \exp((nL-1)H(\frac{n-1}{nL-1})) \leq \exp(nLH(1/L))$. □

The ambient space has $2^{nL} = e^{nL\log 2}$ orthants; the subspace reaches at most $e^{nLH(1/L)}$ of them, and $LH(1/L) = \log L + 1 + O(1/L)$ grows like a logarithm of the depth, not linearly. This is the whole saving.

Remark 4.5 (when the count is attained). If the M central hyperplanes are in general position in a d-dimensional space (every $k \leq d$ of the defining functionals independent), the recurrence (18) is an equality and the region count is exactly $2\sum_{j<d}\binom{M-1}{j}$: removing the last hyperplane leaves an arrangement in general position whose restriction to the removed hyperplane is again in general position, so every region it cuts is split in two and the count is additive. For the all-active mask this happens almost surely (section 10), which is how the exact realizability probability of that mask is computed and checked in section 9.

4.3 The realizable masks: the exact orbit identity

Proposition 4.6 (weighted gauge–orthant identity)**.** *Fix a formal mask sequence D and let $F = F(W_1, \ldots, W_L) \geq 0$ be measurable and gauge invariant. Then*

$$\mathbb{E}\big[F\,\mathbf{1}_{\{R_D \neq \varnothing\}}\big] = \mathbb{E}\Big[F\,\frac{r(\operatorname{im} A_D)}{2^{nL}}\Big]. \tag{20}$$

Consequently

$$\mathbb{E}\big[F\,\mathbf{1}_{\{R_D \neq \varnothing\}}\big] \leq q_{n,L}\,\mathbb{E}F. \tag{21}$$

Proof. By (13), $\mathbf{1}_{\{R_D \neq \varnothing\}} = \mathbf{1}_{\{\operatorname{im} A_D(W) \cap O_D \neq \varnothing\}}$. Fix one $\Sigma \in \mathcal{G}$. The transformed weights have the same joint law as W, so

$$\mathbb{E}\big[F(W)\mathbf{1}_{\{\operatorname{im} A_D(W)\cap O_D \neq \varnothing\}}\big] = \mathbb{E}\big[F(W^\Sigma)\mathbf{1}_{\{\operatorname{im} A_D(W^\Sigma)\cap O_D \neq \varnothing\}}\big].$$

Gauge invariance gives $F(W^\Sigma) = F(W)$, and gauge covariance gives the identity $\operatorname{im} A_D(W^\Sigma) = S_\Sigma \operatorname{im} A_D(W)$. Hence

$$\mathbb{E}\big[F\mathbf{1}_{\{R_D \neq \varnothing\}}\big] = \mathbb{E}\big[F(W)\mathbf{1}_{\{S_\Sigma \operatorname{im} A_D(W)\cap O_D \neq \varnothing\}}\big] \qquad \text{for every } \Sigma \in \mathcal{G}.$$

Average over the 2^{nL} elements of the group:

$$\mathbb{E}\big[F\mathbf{1}_{\{R_D \neq \varnothing\}}\big] = \mathbb{E}\Big[F(W)\,\frac{1}{2^{nL}}\sum_{\Sigma\in\mathcal{G}}\mathbf{1}_{\{S_\Sigma U \cap O_D \neq \varnothing\}}\Big], \qquad U = \operatorname{im} A_D(W). \tag{22}$$

Now fix the realized U. Since $S_\Sigma^{-1} = S_\Sigma$, $S_\Sigma U \cap O_D \neq \varnothing$ if and only if $U \cap S_\Sigma O_D \neq \varnothing$. As Σ ranges over the group, $S_\Sigma O_D$ ranges over all 2^{nL} open orthants exactly once: coordinate by coordinate, the fixed sign of O_D is multiplied by an independently chosen sign, and every sign vector arises from exactly one Σ. Therefore

$$\sum_{\Sigma\in\mathcal{G}}\mathbf{1}_{\{S_\Sigma U \cap O_D \neq \varnothing\}} = \#\{\text{orthants meeting } U\} = r(U),$$

and (22) is (20). Finally $r(\operatorname{im} A_D) \leq C_{n,L}$ by Lemma 4.3, and $F \geq 0$, so the right side of (20) is at most $(C_{n,L}/2^{nL})\mathbb{E}F$. □

Read the identity once more. On the left is the realizability of one formal mask, weighted by any gauge-invariant functional. On the right the realizability event has disappeared; in its place stands a deterministic-looking ratio, the number of orthants the random subspace $\operatorname{im} A_D$ meets divided by the number available. The identity is exact. It does not say that realizability and F are independent: the random count $r(\operatorname{im} A_D)$ may well be correlated with F. What produces the useful inequality is that the count is bounded by a deterministic number, Lemma 4.3. The average over the gauge orbit is the only place the sign symmetry is used, and it is used exactly, not approximately.

This is also why the argument is unconditional: no input is chosen, no activation pattern is conditioned on, no witness input is constructed, and the law of the weights is never tilted. A mask is fixed on paper, a symmetry is averaged, and then all masks are summed.

4.4 The bridge to an independent uniform mask

Let D now be random, independent of the weights, and uniform on the 2^{nL} formal mask sequences.

Corollary 4.7 (unconditional bridge)**.** *For every $t \geq 0$ and every $r > 0$,*

$$\mathbb{P}\big(K_{n,L}(\alpha) > t\big) \leq C_{n,L}\,\mathbb{P}\big(\|J_D\|_{2\to 2} > t\big), \tag{23}$$

$$\mathbb{E}\big[K_{n,L}(\alpha)^r\big] \leq C_{n,L}\,\mathbb{E}\,\|J_D\|_{2\to 2}^r\,. \tag{24}$$

Proof. By Lemma 3.5 and the union bound, $\mathbb{P}(K > t) \leq \sum_D \mathbb{P}(R_D \neq \varnothing, \|J_D\|_{2\to 2} > t)$. For fixed D apply Proposition 4.6 with $F = \mathbf{1}_{\{\|J_D\|_{2\to 2} > t\}}$, which is nonnegative and gauge invariant: $\mathbb{P}(R_D \neq \varnothing, \|J_D\|_{2\to 2} > t) \leq q_{n,L}\mathbb{P}(\|J_D\|_{2\to 2} > t)$. Summing over the 2^{nL} fixed masks, $\sum_D q_{n,L}\mathbb{P}(\|J_D\|_{2\to 2} > t) = C_{n,L}\,2^{-nL}\sum_D \mathbb{P}(\|J_D\|_{2\to 2} > t) = C_{n,L}\mathbb{P}(\|J_D\|_{2\to 2} > t)$ with D uniform and independent. For the moment, (14) gives pointwise $K^r \leq \sum_D \mathbf{1}_{\{R_D \neq \varnothing\}}\,\|J_D\|_{2\to 2}^r$; take expectations and apply Proposition 4.6 with $F = \|J_D\|_{2\to 2}^r$. □

The left side of (23) is the full adaptive supremum over every input. The right side is the Jacobian of a mask chosen by a coin toss, independent of the weights. The entire cost of passing from one to the other is the factor $C_{n,L}$, of exponential size $e^{nLH(1/L)}$ against the 2^{nL} masks that a naive union bound would have paid for.

4.5 Testing the identity

An exact identity can be tested, and (20) was, at width two. In the plane the sign pattern of $A_D s$ is constant on each arc between consecutive boundary directions. The initial rounded-angle enumeration was subsequently replaced by a complete replay using the exact row dependencies of the fixed masks. Section 9 describes its certified represented-weight geometry and clipped-norm error bound. Table 3 reports the corrected paired Monte Carlo test over the weights for $F \equiv 1$, which turns (20) into

$$\mathbb{P}(R_D \neq \varnothing) = \mathbb{E}\, r(\operatorname{im} A_D)/2^{nL},$$

and for the bounded functional $F = \min(\|J_D\|_{2\to 2}\,, 1)$. The initial calculations illustrate two limitations of numerical checks. Sweeping eight thousand directions round the circle instead of enumerating the arcs missed thin arcs and pulled both sides down by about six percent, consistently; sampling a set of arcs is not enumerating it, and the error appeared as a bias rather than noise. Testing with $F = \|J_D\|_{2\to 2}^2$ produced a paired difference two and a half estimated standard errors from zero at depth three. That observation alone establishes neither a failure of the identity nor undercoverage by the standard-error estimate. An unbounded functional can be sensitive to rare large draws, so a bounded functional provides a useful additional check. The identity holds for every nonnegative gauge-invariant F; with the bounded choices used here, every ratio is within two percent of one and every paired z-score within 1.3 of zero.

5 The machine: nets, exact moments, and one scalar variable

Corollary 4.7 leaves a Jacobian J_D with a mask independent of the weights. This section bounds its operator norm by a finite net, computes the moments of $\|J_D v\|$ for a fixed v exactly, and reduces everything to one scalar random variable.

5.1 A net of the sphere

Definition 5.1. A finite set $\mathcal{N} \subseteq S^{n-1}$ is an ε-net if for every $x \in S^{n-1}$ there is $v \in \mathcal{N}$ with $\|x - v\| \le \varepsilon$.

Lemma 5.2. *There is a $\frac{1}{2}$-net $\mathcal{N} \subseteq S^{n-1}$ with $|\mathcal{N}| \le 5^n$, and for every linear $T : \mathbb{R}^n \to \mathbb{R}^m$,*

$$\|T\|_{2\to 2} \le 2 \max_{v\in\mathcal{N}} \|Tv\| . \tag{25}$$

Proof. Take a maximal $\frac{1}{2}$-separated subset $\mathcal{N}$ of the sphere: distinct $u, v \in \mathcal{N}$ have $\|u - v\| > \frac{1}{2}$ and no point of the sphere can be added. Maximality gives the net property, for a point farther than $\frac{1}{2}$ from all of $\mathcal{N}$ could be added. The balls of radius $\frac{1}{4}$ about the points of $\mathcal{N}$ are disjoint, since a common point z would give $\|u - v\| \le \|u - z\| + \|z - v\| \le \frac{1}{2}$; and each lies in the ball of radius $\frac{5}{4}$ about the origin, since $\|z\| \le \|z - v\| + \|v\| \le \frac{1}{4} + 1$. Comparing volumes, which scale as the n-th power of the radius, $|\mathcal{N}|\ (\frac{1}{4})^n \le (\frac{5}{4})^n$, so $|\mathcal{N}| \le 5^n$.

For (25), let x be a unit vector and $v \in \mathcal{N}$ with $\|x - v\| \le \frac{1}{2}$. Then $\|Tx\| \le \|Tv\| + \|T(x - v)\| \le \max_{u\in\mathcal{N}} \|Tu\| + \frac{1}{2} \|T\|_{2\to 2}$. Take the supremum over x and move the last term to the left. □

Raising (25) to the power $2p$ and using $\max_j a_j \le \sum_j a_j$ for nonnegative a_j,

$$\|T\|_{2\to 2}^{2p} \le 4^p \sum_{v\in\mathcal{N}} \|Tv\|^{2p} \qquad (p > 0). \tag{26}$$

Only a net of input directions is needed, because $\|Tv\|$ is a vector norm; a bilinear estimate would need a second net, and the extra factor is unnecessary here. The net never looks for activation regions: it is applied to the fixed linear map J_D, so an arbitrarily narrow realizable cone is never missed.

5.2 Exact moments of $\|J_D v\|$ for a fixed v

Fix a formal mask sequence D and let $S_\ell = \{i : (D_\ell)_{ii} = 1\}$ with $m_\ell = |S_\ell|$ active coordinates. Let $R_\ell : \mathbb{R}^n \to \mathbb{R}^{m_\ell}$ keep exactly the coordinates in S_ℓ; its transpose inserts a vector into those coordinates with zeros elsewhere, so $D_\ell = R_\ell^T R_\ell$, $R_\ell R_\ell^T = I_{m_\ell}$, and $\left\|R_\ell^T y\right\| = \|y\|$. With $R_0 = I_n$, $m_0 = n$, define the active blocks

$$G_1 = R_1 W_1, \qquad G_\ell = R_\ell W_\ell R_{\ell-1}^T \quad (2 \le \ell \le L),$$

of size $m_\ell \times m_{\ell-1}$, obtained by selecting rows and columns of W_ℓ; so all their entries are independent $\mathcal{N}(0, \alpha/n)$ and $G_1, \dots, G_L$ are independent. Inserting $D_\ell = R_\ell^T R_\ell$ into (10),

$$J_D = R_L^T (R_L W_L R_{L-1}^T)(R_{L-1} W_{L-1} R_{L-2}^T) \cdots (R_1 W_1) = R_L^T G_L G_{L-1} \cdots G_1,$$

and since R_L^T preserves norms,

$$\|J_D v\| = \|G_L G_{L-1} \cdots G_1 v\| . \tag{27}$$

If some $m_\ell = 0$ the block has no rows and the product is zero; the formulas below remain valid with $\chi_0^2 = 0$.

One Gaussian block acting on one vector: let G be $m \times k$ with independent $\mathcal{N}(0, \alpha/n)$ entries and $y \in \mathbb{R}^k$ fixed. Row i of Gy is $\sum_j G_{ij} y_j$, centred Gaussian with variance $\frac{\alpha}{n} \|y\|^2$; different rows use disjoint entries, so the coordinates are independent. Hence $Gy \stackrel{\mathrm{d}}{=} \sqrt{\alpha/n}\, \|y\|\, Z$ with $Z \sim \mathcal{N}(0, I_m)$, and

$$\|Gy\|^2 \stackrel{\mathrm{d}}{=} \frac{\alpha}{n} \|y\|^2 \chi^2_m. \tag{28}$$

Lemma 5.3 (exact radial factors)**.** *For every fixed unit vector $v \in \mathbb{R}^n$ and every $p > 0$,*

$$\mathbb{E}\, \|J_D v\|^{2p} = \prod_{\ell=1}^{L} \mathbb{E}\Big(\frac{\alpha}{n} \chi^2_{m_\ell}\Big)^p. \tag{29}$$

Proof. Set $v_0 = v$ and $v_\ell = G_\ell v_{\ell-1}$, so $v_L = G_L \cdots G_1 v$ and $\|J_D v\| = \|v_L\|$ by (27). Conditional on $v_{\ell-1}$, (28) gives $\|v_\ell\|^2 \stackrel{\mathrm{d}}{=} \|v_{\ell-1}\|^2 \frac{\alpha}{n} \chi^2_{m_\ell}$, so $\mathbb{E}[\|v_\ell\|^{2p} \mid v_{\ell-1}] = \|v_{\ell-1}\|^{2p}\, \mathbb{E}(\frac{\alpha}{n} \chi^2_{m_\ell})^p$, the factor being deterministic once m_ℓ is fixed. Take expectations (the tower property) and iterate from $\ell = L$ down to 1; $\|v_0\| = 1$. □

The vectors $v_1, \ldots, v_L$ are not independent, and no such claim is made. The product formula rests only on the conditional identity and on the independence of the new block G_ℓ from everything before it. The expectation depends on D only through the active counts $m_1, \ldots, m_L$ and not on the direction v.

5.3 The scalar mixture

Let $B_1, \ldots, B_n$ be independent Bernoulli($\frac{1}{2}$) and $Z_1, \ldots, Z_n$ independent standard Gaussians, independent of the B_i, and put

$$X_n^{(\alpha)} = \frac{\alpha}{n} \sum_{i=1}^{n} B_i Z_i^2. \tag{30}$$

Conditional on $M_n = \sum_i B_i = m$, exactly m Gaussian squares appear, so $\sum_i B_i Z_i^2 \mid \{M_n = m\} \stackrel{\mathrm{d}}{=} \chi^2_m$, and $M_n \sim \mathrm{Bin}(n, \frac{1}{2})$. Therefore, for every $p > 0$,

$$\mathbb{E}\big[(X_n^{(\alpha)})^p\big] = 2^{-n} \sum_{m=0}^{n} \binom{n}{m} \mathbb{E}\Big(\frac{\alpha}{n} \chi^2_m\Big)^p. \tag{31}$$

This is exactly the sum that appears when the fixed-mask formula (29) is averaged over a uniform mask.

5.4 The master moment inequality

Proposition 5.4 (master inequality)**.** *For every $n, L \geq 1$, $\alpha > 0$ and $p > 0$,*

$$\mathbb{E}\big[K_{n,L}(\alpha)^{2p}\big] \leq C_{n,L}\, 5^n\, 4^p \Big(\mathbb{E}\big[(X_n^{(\alpha)})^p\big]\Big)^L, \tag{32}$$

and consequently

$$\mathbb{P}\big(K_{n,L}(\alpha) > 1\big) \leq C_{n,L}\, 5^n\, 4^p \Big(\mathbb{E}\big[(X_n^{(\alpha)})^p\big]\Big)^L. \tag{33}$$

Proof. Fix the net $\mathcal{N}$ of Lemma 5.2. By (14), pointwise $K^{2p} \leq \sum_D \mathbf{1}_{\{R_D \neq \varnothing\}} \|J_D\|^{2p}_{2\to2}$, and by (26) applied to each J_D,

$$K_{n,L}(\alpha)^{2p} \leq 4^p \sum_D \sum_{v \in \mathcal{N}} \mathbf{1}_{\{R_D \neq \varnothing\}} \|J_D v\|^{2p}.$$

For fixed D and v the functional $\|J_D(W)v\|^{2p}$ is nonnegative and gauge invariant, so Proposition 4.6 gives $\mathbb{E}[\mathbf{1}_{\{R_D\neq\varnothing\}}\,\|J_Dv\|^{2p}] \le (C_{n,L}/2^{nL})\mathbb{E}\,\|J_Dv\|^{2p}$. Hence

$$\mathbb{E}K_{n,L}(\alpha)^{2p} \le 4^p\frac{C_{n,L}}{2^{nL}}\sum_D\sum_{v\in\mathcal{N}}\mathbb{E}\,\|J_Dv\|^{2p} \le 4^p5^n\frac{C_{n,L}}{2^{nL}}\sum_D\mathbb{E}\,\|J_Dv\|^{2p}\,,$$

using that the expectation does not depend on v (Lemma 5.3) and $|\mathcal{N}|\le 5^n$. Group the masks by their active counts: there are $\binom{n}{m}$ diagonal masks with m ones at one layer, so by (29)

$$\sum_D\mathbb{E}\,\|J_Dv\|^{2p} = \prod_{\ell=1}^{L}\Big[\sum_{m=0}^{n}\binom{n}{m}\mathbb{E}\Big(\frac{\alpha}{n}\chi^2_m\Big)^p\Big] = \Big[2^n\,\mathbb{E}(X_n^{(\alpha)})^p\Big]^L$$

by (31). The factor 2^{nL} cancels exactly, which proves (32). On the event $\{K>1\}$ one has $K^{2p}>1$, so Markov's inequality gives (33). $\square$

The four factors and their sources:

factor	source
$C_{n,L}$	the orthants an at-most-n-dimensional stacked subspace can meet
5^n	the size of a $\frac{1}{2}$-net of the input sphere
4^p	$\|T\|_{2\to2}\le 2\max_{v\in\mathcal{N}}\|Tv\|$, raised to the power $2p$
$(\mathbb{E}X_n^p)^L$	one exact radial moment for each of the L independent layers

The proof will choose p proportional to n. Then every factor has an exponential rate in n, and the scalar moment's rate, which is negative when $\alpha<2$, is repeated L times.

5.5 The finite-width scalar moment bound

Lemma 5.5. *Let $Y=BZ^2$ with $B\sim\mathrm{Bernoulli}(\frac{1}{2})$ and $Z\sim\mathcal{N}(0,1)$ independent, and $S_n=\sum_{i=1}^n B_iZ_i^2$ a sum of n independent copies. For $t<\frac{1}{2}$,*

$$\mathbb{E}e^{tY} = \frac{1+(1-2t)^{-1/2}}{2},\qquad \mathbb{E}e^{tS_n} = \Big[\frac{1+(1-2t)^{-1/2}}{2}\Big]^n. \tag{34}$$

Also $\mathbb{E}Y=\frac{1}{2}$, $\mathbb{E}Y^2=\frac{3}{2}$, $\mathrm{Var}(Y)=\frac{5}{4}$.

Proof. $\mathbb{E}e^{tY}=\frac{1}{2}e^0+\frac{1}{2}\mathbb{E}e^{tZ^2}$ and (7); independence gives the product. $\mathbb{E}Y=\mathbb{E}B\,\mathbb{E}Z^2=\frac{1}{2}$, $\mathbb{E}Y^2=\mathbb{E}B\,\mathbb{E}Z^4=\frac{3}{2}$. $\square$

Lemma 5.6 (optimized finite-width scalar moment bound)**.** *For every $n\ge1$, $\alpha>0$ and $\theta>0$, the function in (4) satisfies*

$$\mathbb{E}\big[(X_n^{(\alpha)})^{\theta n}\big]\le e^{ng_\alpha(\theta)}. \tag{35}$$

It extends continuously to $g_\alpha(0)=0$, and

$$g'_\alpha(\theta) = \log\frac{\alpha}{z_\theta^2(1+z_\theta)}, \tag{36}$$

$$g_\alpha(\theta) = \theta\log\frac{\alpha}{2}+\frac{5}{2}\theta^2+O(\theta^3). \tag{37}$$

as $\theta\downarrow0$, with a remainder independent of α.

Proof. Write $\lambda(t) = \log[(1 + (1 - 2t)^{-1/2})/2]$ for $0 < t < 1/2$. Set $p = \theta n$ in Lemma 2.4 and apply it to S_n. Since $X_n^{(\alpha)} = \alpha S_n/n$, (34) gives, for every such t,

$$\frac{1}{n} \log \mathbb{E}(X_n^{(\alpha)})^{\theta n} \leq \theta \log \frac{\alpha\theta}{et} + \lambda(t).$$

The right side has second derivative $\theta/t^2 + \lambda''(t) > 0$ and tends to $+\infty$ at both endpoints, so its unique stationary point is its global minimum. Put $z = \sqrt{1 - 2t}$. Direct differentiation gives

$$\lambda'(t) = \frac{1}{z^2(1 + z)}, \qquad t\lambda'(t) = \frac{1 - z}{2z^2}.$$

The stationary equation $t\lambda'(t) = \theta$ therefore has solution $z = z_\theta$ from (4). At this point $\theta/t = 1/[z_\theta^2(1 + z_\theta)]$, and substitution gives (35).

Differentiate the minimized expression in θ. Its derivative in t vanishes at the minimizing t, so the chain rule leaves $g'_\alpha(\theta) = \log(\alpha\theta/t)$, the derivative in (36). Since $z_\theta = 1 - 2\theta + O(\theta^2)$, this derivative is $\log(\alpha/2) + 5\theta + O(\theta^2)$. The defining formula tends to zero at $\theta = 0$; integration gives the expansion. All dependence on α sits in the exact linear term $\theta \log \alpha$. $\square$

The derivative at zero decides the qualitative threshold: if $\alpha < 2$ then $g'_\alpha(0+) < 0$ and g_α is negative for all small positive θ. At $\alpha = 2$ the linear term vanishes. The coefficient $\frac{5}{2}$ is the normalized variance $\mathrm{Var}(Y)/(2(\mathbb{E}Y)^2) = \frac{5/4}{2 \cdot 1/4}$ of one masked Gaussian square; it is where the constant $\sqrt{10}$ of the rate comes from. Section 10 shows that the entire function g_α, not only its expansion, is the exact limiting moment exponent.

6 The theorems

6.1 The finite-width theorem

Proof of Theorem 1.1. Take $p = \theta n$ in Proposition 5.4. The first inequality in (5) is Markov's inequality on the event $\{K > 1\}$, as in the proof of (33). For the second, Lemma 5.6 gives $\mathbb{E}(X_n^{(\alpha)})^{\theta n} \leq e^{n g_\alpha(\theta)}$, so

$$\mathbb{E}K_{n,L}(\alpha)^{2\theta n} \leq C_{n,L}\, 5^n\, 4^{\theta n}\, e^{nLg_\alpha(\theta)},$$

and Lemma 4.4 bounds $C_{n,L} \leq 2e^{nLH(1/L)}$ for $L \geq 2$. Multiplying out,

$$C_{n,L} 5^n 4^{\theta n} e^{nLg_\alpha(\theta)} \leq 2 \exp\bigl(n[LH(1/L) + \log 5 + \theta \log 4 + Lg_\alpha(\theta)]\bigr) = 2e^{nB_L(\alpha,\theta)}.$$

For $L = 1$, $C_{n,1} = 2^n = 2 \cdot 2^{n-1} \leq 2e^{n \log 2}$ and the same computation applies with $\log 2$ in place of $LH(1/L)$. $\square$

Corollary 1.2 is the theorem with the best θ. The function $-B_L(\alpha, \theta)$ is continuous for $\theta > 0$, tends to $-LH(1/L) - \log 5 < 0$ at zero, and tends to $-\infty$ at infinity. The last claim follows from (36), since $z_\theta \to 0$ and hence $g'_\alpha(\theta) \to +\infty$. Thus whenever its supremum is positive it is attained at an interior θ.

The optimizer is explicit up to a strictly increasing cubic. If $\alpha 4^{1/L} < 2$, there is a unique $z \in (0, 1)$ with

$$z^2(1 + z) = \alpha 4^{1/L}, \qquad \theta = \frac{1 - z}{2z^2}. \tag{38}$$

It minimizes B_L: by (36), $\partial_\theta B_L = \log 4 + L \log[\alpha/(z_\theta^2(1 + z_\theta))]$ is strictly increasing, negative at zero and tends to infinity. At the optimizer,

$$\inf_{\theta > 0} B_L(\alpha, \theta) = LH(1/L) + \log 5 + L\left[-\frac{1 - z}{2z^2} + \log \frac{1 + z}{2z}\right]. \tag{39}$$

If $\alpha 4^{1/L} \geq 2$, the infimum is the positive boundary value $LH(1/L) + \log 5$, so this budget gives no contraction certificate.

6.2 The exponent budget

Written as a rate per unit width, the bound reads

$$\frac{1}{n}\log\mathbb{P}\big(K_{n,L}(\alpha)>1\big)\le\frac{\log 2}{n}+\underbrace{LH(1/L)}_{\text{realizable patterns}}+\underbrace{\log 5}_{\text{sphere net}}+\underbrace{\theta\log 4}_{\text{net to norm}}+\underbrace{Lg_\alpha(\theta)}_{\text{radial moment, } L \text{ layers}}.$$

Three of the four terms do not move once L is fixed, and the first grows only like $\log L$: by Lemma 6.1 below, $LH(1/L)=\log L+1+O(1/L)$. The last term is L times a number that is negative whenever $\alpha<2$ and θ is small. A negative multiple of L eventually beats a logarithm of L; the depth at which it does is $L_0(\alpha)$. This is why the upper bound succeeds at large depth and only there: the realizability entropy costs order $\log L$ per unit width, against the $nL\log 2$ that a union bound over all 2^{nL} masks would cost, and the radial contraction is order L.

Lemma 6.1. *$LH(1/L)=\log L-(L-1)\log(1-1/L)=\log L+1+O(1/L)$ as $L\to\infty$, and $LH(1/L)\le\log L+1$ for every $L\ge 2$.*

Proof. $LH(1/L)=-\log(1/L)-(L-1)\log(1-1/L)$. Since $-\log(1-x)=x+x^2/2+\cdots\ge x$, $-(L-1)\log(1-1/L)\ge(L-1)/L$, and since $-\log(1-x)\le x/(1-x)$, $-(L-1)\log(1-1/L)\le(L-1)\cdot\frac{1/L}{1-1/L}=1$. So $\log L+1-1/L\le LH(1/L)\le\log L+1$. $\square$

6.3 Fixed depth, growing width; the depth threshold

Proof of Theorem 1.4(i). Fix $\alpha<2$. By (37), $g'_\alpha(0+)=\log(\alpha/2)<0$, so there is $\theta>0$, depending on α but not on n or L, with $g_\alpha(\theta)=-\gamma<0$. By Lemma 6.1 the budget satisfies

$$B_L(\alpha,\theta)\le\log L+1+\log 5+\theta\log 4-L\gamma,$$

which is negative for all $L\ge L_0(\alpha)$, a linear function eventually dominating a logarithm. For such L, Theorem 1.1 gives $\mathbb{P}(K_{n,L}(\alpha)>1)\le 2e^{-c_{\alpha,L}n}$ with $c_{\alpha,L}=-B_L(\alpha,\theta)>0$, for every width n. $\square$

The smallest depth at which the exact budget $\inf_{\theta>0}B_L(\alpha,\theta)$ turns negative is a computable number; for $\alpha=1$ it is 119, for $\alpha=1.9$ it is 50 026 (section 9). At depths below it the bound says nothing, and at depth one it cannot: $\alpha_1=1/4$.

6.4 The other side

Lemma 6.2 (one fixed input)**.** *Fix L and a deterministic $s\ne 0$. Conditional on $x_{\ell-1}(s)\ne 0$,*

$$\frac{\|x_\ell(s)\|^2}{\|x_{\ell-1}(s)\|^2}\overset{\mathrm{d}}{=}\frac{\alpha}{n}\sum_{i=1}^{n}B_{\ell i}Z_{\ell i}^2=X_n^{(\alpha)},\tag{40}$$

with the pairs $(B_{\ell i},Z_{\ell i})$ independent across i. Also, for every nonzero s, whether or not the network is differentiable there,

$$K_{n,L}(\alpha)\ge\frac{\left\|F_{n,L}^{(\alpha)}(s)\right\|}{\|s\|}.\tag{41}$$

Proof. Given the past through layer $\ell-1$, $x_{\ell-1}(s)$ is fixed and W_ℓ independent of it, so the coordinates of $W_\ell x_{\ell-1}(s)$ are independent centred Gaussians of variance $\frac{\alpha}{n}\|x_{\ell-1}(s)\|^2$; ReLU keeps each with probability $\frac{1}{2}$, independently of its magnitude (a centred Gaussian's sign and absolute value are independent), which is (40). For (41), $F(0)=0$ by (6), so $K\ge\|F(s)-F(0)\|\,/\,\|s-0\|$. $\square$

Proof of Theorem 1.3. Fix $s = e_1$ and write $r_\ell = \|x_\ell\|^2 / \|x_{\ell-1}\|^2$ when $x_{\ell-1} \neq 0$. If no layer dies and every $r_\ell > 1$, then $\|x_L\|^2 / \|s\|^2 = \prod_\ell r_\ell > 1$ and $K_{n,L}(\alpha) > 1$ by (41). So

$$\{K_{n,L}(\alpha) \le 1\} \subseteq \bigcup_{\ell=1}^{L} \{x_{\ell-1} \neq 0,\ \tfrac{\alpha}{n} S_n^{(\ell)} \le 1\} \cup \{\text{some layer dies}\},$$

where $S_n^{(\ell)} = \sum_i B_{\ell i} Z_{\ell i}^2$ is the sum in (40). A layer dies exactly when all n of its Bernoulli variables are zero, in which case $S_n^{(\ell)} = 0 \le n/\alpha$; so the dying event is contained in the same union. By the union bound and (40), $\mathbb{P}(K \le 1) \le L\,\mathbb{P}(S_n \le n/\alpha)$ with S_n as in Lemma 5.5. Since $\mathbb{E}S_n = n/2$, $\operatorname{Var} S_n = 5n/4$ and $n/\alpha < n/2$ for $\alpha > 2$, Chebyshev's inequality gives

$$\mathbb{P}\Big(S_n \le \frac{n}{\alpha}\Big) \le \mathbb{P}\Big(\big|S_n - \tfrac{n}{2}\big| \ge \frac{n}{2} - \frac{n}{\alpha}\Big) \le \frac{5n/4}{n^2(\frac{1}{2} - \frac{1}{\alpha})^2} = \frac{5\alpha^2}{n(\alpha-2)^2}.$$

Multiply by L. □

Theorem 1.4(ii) follows, since the bound tends to zero as $n \to \infty$ at fixed L and $\alpha > 2$. A Chernoff bound with the moment-generating function (34) gives an exponentially small bound in place of $1/n$; the weaker form is enough for the theorem and has an explicit constant.

Remark 6.3. The reverse direction needs no largest singular value, no rare activation pattern, and no global search. The ordinary forward direction of one fixed input already expands, because each layer multiplies the squared norm by an average of $\alpha/2 > 1$ and the fluctuations are of order $n^{-1/2}$.

6.5 The depth limit

Proof of Theorem 1.4(iii). Part (ii) says that no $\alpha > 2$ belongs to the set in (3) for any L, so $\alpha_L \le 2$ for every L. Fix $\alpha < 2$; by part (i), for every $L \ge L_0(\alpha)$, $\mathbb{P}(K_{n,L}(\alpha) \le 1) \to 1$, so α belongs to the set and $\alpha_L \ge \alpha$ for all such L. Hence $\liminf_L \alpha_L \ge \alpha$ for every $\alpha < 2$, so $\liminf_L \alpha_L \ge 2 \ge \limsup_L \alpha_L$. □

Reading the statement carefully: for every $\alpha < 2$ there is a depth $L_0(\alpha)$, inside the choice of α, such that for every $L \ge L_0(\alpha)$ the probability statement holds as $n \to \infty$. The sufficient depth supplied by this budget diverges as $\alpha \uparrow 2$. This does not prove that a diverging depth is necessary for the network itself.

6.6 The quantitative rate

Write $\alpha = 2e^{-\delta}$. Then $\theta \log(\alpha/2) = -\delta\theta$, and by (37)

$$g_{2e^{-\delta}}(\theta) = -\delta\theta + \frac{5}{2}\theta^2 + O(\theta^3), \tag{42}$$

with a remainder uniform in δ near zero, since all dependence on α is in the exact linear term. Ignoring the remainder, $-\delta\theta + \frac{5}{2}\theta^2$ is minimized at $\theta = \delta/5$ with value $-\delta^2/10$; so L layers produce a negative exponent of about $-L\delta^2/10$, against a geometric cost of about $\log L$. Balancing, $L\delta^2/10 \approx \log L$, or $\delta \approx \sqrt{10 \log L / L}$. The proof makes this precise.

Proof of Theorem 1.4(iv). Fix a constant $A > 1 + \log 5$ and for large L put

$$\delta_L = \sqrt{\frac{10(\log L + A)}{L}}, \qquad \alpha(L) = 2e^{-\delta_L}, \qquad \theta_L = \frac{\delta_L}{5}.$$

Since $\delta_L \to 0$, $\theta_L > 0$ for all large L. Evaluate the budget of Theorem 1.1 at $(\alpha(L), \theta_L)$, with L fixed and n free:

$$\begin{aligned} B_L(\alpha(L), \theta_L) &= LH(1/L) + \log 5 + \theta_L \log 4 + L g_{\alpha(L)}(\theta_L) \\ &= \log L + 1 + O(1/L) + \log 5 + o(1) + L\Big[-\frac{\delta_L^2}{5} + \frac{5}{2}\cdot\frac{\delta_L^2}{25} + O(\delta_L^3)\Big] \\ &= \log L + 1 + \log 5 - \frac{L\delta_L^2}{10} + O(L\delta_L^3) + o(1). \end{aligned}$$

By definition $L\delta_L^2/10 = \log L + A$, while $L\delta_L^3 = 10^{3/2}(\log L + A)^{3/2} L^{-1/2} \to 0$. Hence

$$B_L(\alpha(L), \theta_L) = 1 + \log 5 - A + o(1),$$

which is negative for all large L because $A > 1 + \log 5$. For such L, Theorem 1.1 gives $\mathbb{P}(K_{n,L}(\alpha(L)) > 1) \to 0$ as $n \to \infty$, so $\alpha(L)$ belongs to the set defining α_L and

$$\alpha_L \geq \alpha(L) = 2\exp\Big[-\sqrt{\frac{10(\log L + A)}{L}}\Big] = 2\exp\Big[-(\sqrt{10} + o(1))\sqrt{\frac{\log L}{L}}\Big],$$

the last step because $\sqrt{1 + A/\log L} = 1 + o(1)$. □

The rate is the balance between one number that comes from geometry, the entropy $\log L$ of the realizable patterns, and one that comes from probability, the curvature $\frac{5}{2}$ of the moment exponent at the critical variance. The asymptotic form drops the $O(1/L)$, the $\theta_L \log 4$, the $O(L\delta_L^3)$ and the choice of A, and is therefore only the large-L shape of the bound; at a specific depth the number the proof delivers is the threshold of the exact budget,

$$\alpha_L^* = \sup\Big\{\alpha : \inf_{\theta > 0} B_L(\alpha, \theta) < 0\Big\} \leq \alpha_L, \tag{43}$$

which section 9 tabulates. At $L = 100$ the exact budget gives 0.954 while the asymptotic formula, read at $L = 100$ with the $o(1)$ dropped, would say 1.01, a number the proof does not deliver there.

6.7 Exact scaling in the variance

Let $G_1, \dots, G_L$ have independent $\mathcal{N}(0, 1/n)$ entries and couple $W_\ell^{(\alpha)} = \sqrt{\alpha}\, G_\ell$. Each layer contributes one factor $\sqrt{\alpha}$, and by homogeneity of ReLU, $F_{n,L}^{(\alpha)}(s) = \alpha^{L/2} F_{n,L}^{(1)}(s)$, hence

$$K_{n,L}(\alpha) = \alpha^{L/2} K_{n,L}(1). \tag{44}$$

The positive scalar changes no activation sign, so every mask is unchanged under the coupling. Stability is therefore monotone in α: the set in (3) is an interval $(0, \alpha_L)$ or $(0, \alpha_L]$, and the word threshold is literal. The identity also says that the whole question is the distribution of $K_{n,L}(1)$, and that $\mathbb{P}(K_{n,L}(\alpha) \leq 1) = \mathbb{P}(K_{n,L}(1) \leq \alpha^{-L/2})$.

6.8 Biases

The global probability bounds extend to independent symmetric biases. Write F_b for the network with $h_\ell(s) = W_\ell x_{\ell-1}(s) + b_\ell$ and $K_{n,L}^{(b)}(\alpha) = \mathrm{Lip}(F_b)$.

Theorem 6.4 (finite width, with biases)**.** *Suppose all nL coordinates of $(b_1, \dots, b_L)$ are mutually independent, each is symmetric about zero, and the entire bias array is independent of the weights. For every $n \geq 1$, $L \geq 2$, $\alpha > 0$ and $\theta > 0$,*

$$\mathbb{P}\big(K_{n,L}^{(b)}(\alpha) > 1\big) \leq 2e^{nB_L(\alpha,\theta)}.$$

For $\alpha > 2$,

$$\mathbb{P}\big(K_{n,L}^{(b)}(\alpha) \le 1\big) \le \frac{5\alpha^2 L}{n(\alpha-2)^2}.$$

No moment or size bound on the biases is required. In particular, independent centred Gaussian biases of arbitrary variances are allowed.

Proof. Affine regions and symmetry. For a formal mask D, the preactivations are $H_\ell^D s + c_\ell^D$, where $c_1^D = b_1$ and $c_\ell^D = W_\ell D_{\ell-1} c_{\ell-1}^D + b_\ell$. The Jacobian is still J_D. The proof of Lemma 3.4 applies to affine pieces, since their constant terms cancel in differences. The same conditional Gaussian row argument as in section 3 shows that every full-dimensional nonconstant piece has strict preactivation signs on its interior, almost surely. If a formal linear part vanishes, every later linear part vanishes as well; such a piece has zero Jacobian. Thus the reduction to nonempty strict regions still holds.

With c_D the stacked offsets, realizability means that $\operatorname{im} A_D + c_D$ meets O_D. Extend the gauge by $b_\ell \mapsto \Sigma_\ell b_\ell$. Induction gives $c_\ell^D(W^\Sigma, b^\Sigma) = \Sigma_\ell c_\ell^D(W,b)$, while Lemma 4.2 gives the same transformation of the linear parts. The assumed independence across *all layers and coordinates* preserves the joint law of (W,b) under every such sign change. Hence, for every nonnegative gauge-invariant functional G, Proposition 4.6 gives

$$\mathbb{E}\big[G\mathbf{1}_{\{R_D\neq\varnothing\}}\big] = \mathbb{E}\Big[G\,\frac{r(\operatorname{im} A_D + c_D)}{2^{nL}}\Big].$$

Counting and moments. An affine subspace of dimension $d \le n$ in $\mathbb{R}^M$ meets at most $\sum_{j=0}^{d}\binom{M}{j}$ open orthants. Restrict the coordinate hyperplanes to that subspace: their regions satisfy the arrangement recurrence of (18), now with boundary values $R(M,0) = R(0,d) = 1$. Pascal's identity gives the stated bound. A coordinate identically zero gives no open orthants and causes no exception. With $M = nL$, Lemma 2.3 gives

$$r(\operatorname{im} A_D + c_D) \le \sum_{j=0}^{n}\binom{nL}{j} \le e^{nLH(1/L)}.$$

The net and radial-moment estimates concern J_D alone. Substituting this count into the proof of Proposition 5.4 and Theorem 1.1 gives the stated upper bound with the same budget B_L.

The expansive side. A deterministic argument handles every bias realization, including paths that die. Let F_0 have the same weights and zero biases. Set $C_0 = 0$ and $C_\ell = \|W_\ell\|_{2\to2}\, C_{\ell-1} + \|b_\ell\|$. The 1-Lipschitz property of ReLU gives, by induction,

$$\sup_s \|F_b(s) - F_0(s)\| \le C_L < \infty.$$

Since $F_0(cs) = cF_0(s)$ for $c > 0$, it follows that $F_b(cs)/c \to F_0(s)$. For any $x \ne y$,

$$K_{n,L}^{(b)}(\alpha) \ge \lim_{c\to\infty} \frac{\|F_b(cx) - F_b(cy)\|}{c\,\|x-y\|} = \frac{\|F_0(x) - F_0(y)\|}{\|x-y\|}.$$

Taking the supremum proves $K_{n,L}^{(b)}(\alpha) \ge K_{n,L}(\alpha)$ for these coupled networks. The second probability bound follows from Theorem 1.3. □

Thus the same sufficient depth and width certificates apply: for each $\alpha < 2$, sufficiently large fixed depth gives exponentially small failure probability as width grows; for $\alpha > 2$, every fixed depth is expansive with probability tending to one. The exact variance-scaling identity (44) remains a bias-free statement. The contraction bound requires the joint bias symmetry above; symmetry within each layer alone does not suffice.

7 One matrix, many steps: the finite-horizon norm law

The network of section 1 uses a fresh matrix at every layer. This section treats the other natural object: one Gaussian matrix applied again and again,

$$x_{t+1} = \phi(Wx_t), \qquad t = 0, 1, 2, \ldots, \qquad W \in \mathbb{R}^{N\times N}, \quad W_{ij} \overset{\text{ind}}{\sim} \mathcal{N}(0, 1/N), \tag{45}$$

started from a unit vector $x_0 \in S^{N-1}$. In the notation of section 1 this is width $n = N$ and variance parameter $\alpha = 1$, with the L independent matrices replaced by a single one. The question is no longer the worst input but the ordinary one: what does the norm of the trajectory do? At $\alpha = 1$ an independent Gaussian layer halves the conditional expected squared norm. The theorem shows that the tied iteration follows the same norm profile for a fixed number of steps with exponentially small failure probability, for every deterministic unit input and outside a set of exponentially small measure for one typical matrix. The proof is due to the same author as the global theorem and is given here in a condensed form; nothing is omitted from the logic, only from the commentary.

7.1 Statements

Theorem 7.1 (deterministic input)**.** *Fix an integer $T \geq 0$ and $\delta > 0$. There are constants $C, c > 0$, depending only on T and δ, such that for every $N \geq 1$ and every deterministic $x_0 \in S^{N-1}$,*

$$\mathbb{P}_W\Big(\max_{0\leq t\leq T} \big|2^t \left\|x_t\right\|^2 - 1\big| > \delta\Big) \leq Ce^{-cN}. \tag{46}$$

Corollary 7.2 (random start independent of the matrix)**.** *If X_0 is a random unit vector independent of W, the same bound holds for $\mathbb{P}_{W,X_0}$.*

For a fixed matrix W let $\mathcal{B}(W) = \{x_0 \in S^{N-1} : \max_{t\leq T} |2^t \left\|x_t\right\|^2 - 1| > \delta\}$ be its set of bad starting points.

Theorem 7.3 (small exceptional set of inputs)**.** *Fix T and δ and let μ_N be a probability measure on S^{N-1}, deterministic or random but independent of W. With the constants of Theorem 7.1, conditionally on μ_N,*

$$\mathbb{P}_W\big(\mu_N(\mathcal{B}(W)) > e^{-cN/2}\big) \leq Ce^{-cN/2}.$$

The one-step case is the warm-up. For deterministic x_0, the coordinates of Wx_0 are independent $\mathcal{N}(0, 1/N)$, so $\left\|x_1\right\|^2 = \frac{1}{N}\sum_i (\xi_i)_+^2$ with ξ_i standard normal, and $\mathbb{E}(\xi_+)^2 = \frac{1}{2}\mathbb{E}\xi^2 = \frac{1}{2}$ by symmetry: the law of large numbers puts $\left\|x_1\right\|^2$ near $\frac{1}{2}$. The difficulty begins at the second step. The vector x_1 depends on W, so Wx_1 is not a Gaussian vector with independent coordinates, and the one-step argument cannot simply be repeated. This is the same entanglement as in section 3, inside a single matrix. The proof handles it by conditioning, and the point of the section is why conditioning is legal here when it was avoided there.

7.2 Tools

Beyond section 2, three estimates are needed. For $Z \sim \mathcal{N}(0, \sigma^2)$ and $t > 0$, Markov's inequality on $e^{\lambda Z}$ with $\lambda = t/\sigma^2$ gives the Gaussian tail

$$\mathbb{P}(|Z| \geq t) \leq 2e^{-t^2/(2\sigma^2)}. \tag{47}$$

Lemma 7.4 (elementary Bernstein inequality)**.** *Let $X_1, \ldots, X_N$ be independent, mean zero, with $\mathbb{E}\exp(|X_i|/K) \leq 2$ for every i. Then for every $u > 0$,*

$$\mathbb{P}\Big(\Big|\frac{1}{N}\sum_{i=1}^N X_i\Big| > u\Big) \leq 2\exp\Big[-N\min\Big\{\frac{u^2}{16K^2}, \frac{u}{8K}\Big\}\Big].$$

Proof. Expanding the exponential, $2 \geq \mathbb{E}e^{|X_i|/K} = \sum_p \mathbb{E}\,|X_i|^p/(p!K^p)$ termwise, so $\mathbb{E}\,|X_i|^p \leq 2p!K^p$. For $|\lambda|\,K \leq \frac{1}{4}$, since $\mathbb{E}X_i = 0$,

$$\mathbb{E}e^{\lambda X_i} = 1 + \sum_{p\geq 2} \frac{\lambda^p \mathbb{E}X_i^p}{p!} \leq 1 + 2\sum_{p\geq 2}(|\lambda|\,K)^p = 1 + \frac{2(|\lambda|\,K)^2}{1-|\lambda|\,K} \leq 1 + 4\lambda^2K^2 \leq e^{4\lambda^2K^2}.$$

By independence and Markov, $\mathbb{P}(\frac{1}{N}\sum X_i > u) \leq \exp(-\lambda N u + 4N\lambda^2K^2)$ for $0 < \lambda \leq 1/(4K)$. If $u \leq 2K$ take $\lambda = u/(8K^2)$, which gives $e^{-Nu^2/(16K^2)}$; if $u > 2K$ take $\lambda = 1/(4K)$, which gives $\exp(-Nu/(4K) + N/4) \leq e^{-Nu/(8K)}$. The lower tail is the same argument for $-X_i$. $\square$

Lemma 7.5 (average squared length of Gaussian vectors)**.** *If $\Gamma_1, \dots, \Gamma_N$ are independent $\mathcal{N}(0, I_d)$ then $\mathbb{P}\big(\frac{1}{N}\sum_i \|\Gamma_i\|^2 > 2d\big) \leq \exp(-\frac{1-\log 2}{2}\,dN)$.*

Proof. $S = \sum_i \|\Gamma_i\|^2$ is a sum of Nd independent squared standard normals, so $\mathbb{E}e^{S/4} = (1-\frac{1}{2})^{-Nd/2} = 2^{Nd/2}$ by (7), and $\mathbb{P}(S > 2Nd) \leq e^{-Nd/2}2^{Nd/2}$. $\square$

7.3 What a query reveals: Gaussian innovations

Write w_i^T for the i-th row of W; the rows are independent $\mathcal{N}(0, I_N/N)$ vectors. A matrix–vector query $g = Wu$ with a unit vector u returns $g_i = \langle w_i, u\rangle$, the coefficient of each row in the direction u and nothing else: $w_i = \langle w_i, u\rangle u + P_{u^\perp} w_i$, and the query determines the first term only. After orthonormal queries $u_1, \dots, u_r$ the revealed part of every row is its projection onto $U_r = \mathrm{span}(u_1, \dots, u_r)$, and the unresolved residual is the projection onto $U_r^\perp$. The identity is pure geometry and holds when the directions are chosen adaptively. The probabilistic claim is that the residual is still Gaussian, and that needs care, because U_r depends on W.

The mechanism in one line: if A, B are independent standard normals and $s(A) \in \{\pm 1\}$ is any measurable function of A alone, then $C = s(A)B$ is standard normal and independent of A, because conditionally on A the sign is fixed and B is symmetric. The rule may depend on the past in any way; it may not look at the fresh variable it is applied to. The innovation theorem is the matrix version.

Lemma 7.6 (one fresh direction inside an unobserved subspace)**.** *Let P be a deterministic orthogonal projection on $\mathbb{R}^N$, u a unit vector with $Pu = 0$, and Z an $N \times N$ matrix with independent $\mathcal{N}(0, 1/N)$ entries. Put $P^+ = P + uu^T$, $\xi = Zu$ and $R = Z(I - P^+)$. Then P^+ is the orthogonal projection onto* $\mathrm{range}(P) \oplus \mathrm{span}(u)$*; $Z(I-P) = \xi u^T + R$; $\xi \sim \mathcal{N}(0, I_N/N)$; ξ and R are independent; and $R \overset{\mathrm{d}}{=} \widetilde{Z}(I - P^+)$ for an independent copy $\widetilde{Z}$ of Z.*

Proof. $(P + uu^T)^2 = P + uu^T$ because $Pu = 0$ and $\|u\| = 1$, and $P + uu^T$ is symmetric, so it is the orthogonal projection onto the stated sum. Since $I - P = uu^T + (I - P^+)$, multiplying by Z gives the identity. The i-th coordinate of ξ is $z_i^T u$ with z_i the i-th row of Z, a centred Gaussian of variance $u^T(I_N/N)u = 1/N$, independent across rows. The i-th row of R is $r_i = (I - P^+)z_i$, jointly Gaussian with $\xi_i = z_i^T u$, with cross-covariance $\mathbb{E}[r_i\xi_i] = (I - P^+)\mathbb{E}[z_i z_i^T]u = \frac{1}{N}(I - P^+)u = 0$ because $P^+u = u$; jointly Gaussian and uncorrelated means independent, and the row pairs are independent across i, so ξ is independent of the whole matrix R. Finally R is a fixed linear image of Z, so replacing Z by an independent copy does not change its law. $\square$

Let $\mathcal{F}_0$ be trivial and $\mathcal{F}_k = \sigma(g_1, \dots, g_k)$; put $U_k = [u_1 \cdots u_k]$, $P_k = U_k U_k^T$ and $M_k = \sum_{j\leq k} g_j u_j^T$, so that $M_k u_a = g_a$ for $a \leq k$: M_k is the revealed action of W on the queried span.

Lemma 7.7 (Gaussian innovations under adaptive orthogonal queries)**.** *Fix $K \le N$. Let $u_1 \in S^{N-1}$ be deterministic and $g_1 = Wu_1$. For $2 \le k \le K$ let u_k be $\mathcal{F}_{k-1}$-measurable with, almost surely, $\|u_k\| = 1$ and $\langle u_k, u_j\rangle = 0$ for $j < k$, and put $g_k = Wu_k$. Then for every $k \le K$,*

$$W \mid \mathcal{F}_k \overset{\mathrm{d}}{=} M_k + \widetilde{W}_k(I - P_k), \tag{48}$$

with $\widetilde{W}_k$ a fresh $\mathcal{N}(0, 1/N)$ matrix independent of $\mathcal{F}_k$ (meaning $\mathbb{E}[\Phi(W) \mid \mathcal{F}_k] = \mathbb{E}_Z\Phi(M_k + Z(I - P_k))$ for bounded Borel Φ, the revealed M_k, P_k held fixed); $g_k \mid \mathcal{F}_{k-1} \sim \mathcal{N}(0, I_N/N)$; and $g_1, \dots, g_K$ are independent $\mathcal{N}(0, I_N/N)$ vectors.

Proof. Induction on k; at $k = 0$ the statement is the law of W. Assume (48) at $k-1$. Conditionally on $\mathcal{F}_{k-1}$, the vectors $g_1, \dots, g_{k-1}$, $u_1, \dots, u_k$ and the matrices M_{k-1}, P_{k-1} are fixed, u_k because it is $\mathcal{F}_{k-1}$-measurable: it may have been chosen by an elaborate rule, but once the past is revealed the rule has produced a known vector. Multiply (48) by u_k: $M_{k-1}u_k = \sum_{j<k} g_j\langle u_j, u_k\rangle = 0$ by orthogonality, and $(I - P_{k-1})u_k = u_k$, so $g_k = Wu_k \overset{\mathrm{d}}{=} Zu_k$ with Z fresh; conditionally on the past u_k is a fixed unit vector, so $g_k \mid \mathcal{F}_{k-1} \sim \mathcal{N}(0, I_N/N)$, a law that does not depend on the past: g_k is independent of $\mathcal{F}_{k-1}$. For the representation at step k apply Lemma 7.6 with $P = P_{k-1}$ and $u = u_k$: $Z(I - P_{k-1}) = \xi u_k^T + R$ with $\xi = Zu_k = g_k$ and $R \overset{\mathrm{d}}{=} \widetilde{Z}(I - P_k)$ independent of ξ. So, conditionally on $\mathcal{F}_{k-1}$, the pair (g_k, W) has the law of $(\xi, M_{k-1} + \xi u_k^T + R)$ with ξ independent of R; conditioning further on $\xi = g_k$ leaves the law of R unchanged and gives $W \mid \mathcal{F}_k \overset{\mathrm{d}}{=} M_{k-1} + g_k u_k^T + \widetilde{Z}(I - P_k) = M_k + \widetilde{Z}(I - P_k)$. Joint independence: for bounded φ_j, $\mathbb{E}\prod_{j\le K}\varphi_j(g_j) = \mathbb{E}[\prod_{j<K}\varphi_j(g_j)\,\mathbb{E}[\varphi_K(g_K) \mid \mathcal{F}_{K-1}]] = \mathbb{E}[\prod_{j<K}\varphi_j(g_j)]\,\mathbb{E}\varphi_K(G)$ with $G \sim \mathcal{N}(0, I_N/N)$, and repeat. $\square$

Three hypotheses, each necessary. Orthogonality: if $u_2 = u_1$ then $g_2 = g_1$; in general $Wu_k = \sum_{j<k}\alpha_j g_j + Wu_k^{\perp}$ and only the orthogonal part is fresh, which is what Gram–Schmidt isolates. The direction may depend only on the past: a rule allowed to inspect the unobserved residual could pick the direction of its largest projection. Gaussianity: for a non-Gaussian isotropic vector, orthogonal projections can be uncorrelated without being independent. And the matrix model matters: for a symmetric Gaussian matrix the rows are coupled and the statement is false as written.

7.4 Gram–Schmidt on the trajectory

Let $V_t = \mathrm{span}(x_0, \dots, x_t)$, of dimension at most $t+1$. Apply Gram–Schmidt to $x_0, x_1, \dots$ in order: $v_1 = x_0$; when a new x_t has a nonzero residual $\rho_t = x_t - \sum_{j\le r}\langle x_t, v_j\rangle v_j$, set $v_{r+1} = \rho_t/\|\rho_t\|$. Each new direction is a unit vector orthogonal to the earlier ones, and it is a measurable function of the images $g_j = Wv_j$ already exposed: $x_1 = \phi(g_1)$, and inductively $Wx_t = \sum_j\langle x_t, v_j\rangle g_j$ so $x_{t+1} = \phi(Wx_t)$, its projection onto V_t, its residual and the normalized residual are all functions of $g_1, \dots, g_r$. So the Gram–Schmidt directions are admissible probes for Lemma 7.7.

One technical point keeps the probe count deterministic. Lemma 7.7 is stated for a fixed number K of probes; the number of directions the trajectory actually creates is random. Process the states through x_T in order: while a state lies in the exposed span, compute its next image using the known action of W; whenever a new residual appears, query its normalized direction. Only after this trajectory processing is complete, pad to $d = T + 1 \le N$ probes by a fixed transcript-measurable rule (the first standard basis vector not yet in the span, projected and normalized). The list has deterministic length and is never conditioned on the number of directions created by the trajectory. This produces exactly d transcript-measurable orthonormal probes $v_1, \dots, v_d$, with $V_T \subseteq \mathrm{span}(v_1, \dots, v_d)$ and every padding vector orthogonal to V_T; by Lemma 7.7, $g_1, \dots, g_d$ are independent $\mathcal{N}(0, I_N/N)$. For $0 \le t \le T$ define the coefficient vector

$a_t = (\langle x_t, v_1\rangle, \dots, \langle x_t, v_d\rangle) \in \mathbb{R}^d$; the padding coordinates are zero, Parseval gives $\|a_t\| = \|x_t\|$, and the driving representation is

$$Wx_t = \sum_{j=1}^{d} (a_t)_j g_j. \tag{49}$$

For each output coordinate i let $\Gamma_i = \sqrt{N}\big((g_1)_i, \dots, (g_d)_i\big) \in \mathbb{R}^d$; the Nd scalars $\sqrt{N}(g_j)_i$ are independent standard normals, so $\Gamma_1, \dots, \Gamma_N$ are independent $\mathcal{N}(0, I_d)$. Taking the i-th coordinate of (49), applying ReLU and homogeneity, squaring and summing,

$$\|x_{t+1}\|^2 = \frac{1}{N}\sum_{i=1}^{N} (a_t \cdot \Gamma_i)_+^2. \tag{50}$$

The whole trajectory is now written in terms of d independent Gaussian vectors and $T+1$ coefficient vectors that live in a space of dimension $d = T+1$, a dimension that does not grow with N. The coefficient vectors are random and depend on the same Gaussians; a concentration estimate for one fixed a would not be enough, and the next lemma is uniform over a ball.

7.5 A uniform law of large numbers in a low dimension

Lemma 7.8 (uniform Gaussian law of large numbers)**.** *Fix d, $0 \le R < \infty$ and $\varepsilon > 0$, and let $\Gamma_1, \dots, \Gamma_N$ be independent $\mathcal{N}(0, I_d)$. There are $C, c > 0$ depending only on d, R, ε such that*

$$\mathbb{P}\Big(\sup_{\|a\|\le R}\Big|\frac{1}{N}\sum_{i=1}^{N}(a\cdot\Gamma_i)_+^2 - \frac{1}{2}\|a\|^2\Big| > \varepsilon\Big) \le Ce^{-cN}.$$

Proof. For $R = 0$ the supremum is zero and the conclusion is immediate. Assume $R > 0$ and write $A_N(a) = \frac{1}{N}\sum_i (a\cdot\Gamma_i)_+^2$. For fixed a, $Z = a\cdot\Gamma$ is $\mathcal{N}(0, \|a\|^2)$ and $\mathbb{E}Z_+^2 = \frac{1}{2}\mathbb{E}Z^2 = \frac{1}{2}\|a\|^2$ by symmetry, so $\mathbb{E}A_N(a) = \frac{1}{2}\|a\|^2$. Concentration for one a with $\|a\| \le R$: $Y_i = (a\cdot\Gamma_i)_+^2$ satisfies $0 \le Y_i \le Z_i^2$ and $\mathbb{E}Y_i \le R^2/2$, so $X_i = Y_i - \mathbb{E}Y_i$ has $|X_i| \le Z_i^2 + R^2/2$, and with $K_R = 32R^2$, by (7), $\mathbb{E}e^{|X_i|/K_R} \le e^{1/64}(1 - 2R^2/(32R^2))^{-1/2} = e^{1/64}(1-\frac{1}{16})^{-1/2} < 2$. Lemma 7.4 with $u = \varepsilon/2$ gives $\mathbb{P}(\big|A_N(a) - \frac{1}{2}\|a\|^2\big| > \varepsilon/2) \le 2e^{-c_1 N}$ with $c_1 = \min\{\varepsilon^2/(64K_R^2), \varepsilon/(16K_R)\}$, which depends on R, ε only. Nearby coefficient vectors: for $\|a\|, \|b\| \le R$ and any Γ, using $|u^2 - v^2| = |u-v|\,|u+v|$, the 1-Lipschitz property of ϕ and Cauchy–Schwarz, $\big|(a\cdot\Gamma)_+^2 - (b\cdot\Gamma)_+^2\big| \le 2R\|a-b\|\,\|\Gamma\|^2$; averaging,

$$|A_N(a) - A_N(b)| \le 2R\|a-b\|\,\frac{1}{N}\sum_i \|\Gamma_i\|^2,$$

and the means differ by at most $R\|a-b\|$. Net: put $\alpha_0 = \min\{R, \varepsilon/(2R(4d+1))\}$. The ball $B_d(R)$ has an α_0-net $\mathcal{N}$ of size $M \le (3R/\alpha_0)^d$ (a maximal α_0-separated set, by the volume argument of Lemma 5.2 with radii $\alpha_0/2$ and $R + \alpha_0/2$), a number independent of N. On the event that every net point concentrates to within $\varepsilon/2$, which fails with probability at most $2Me^{-c_1N}$, and the event $\frac{1}{N}\sum\|\Gamma_i\|^2 \le 2d$ of Lemma 7.5, which fails with probability at most e^{-c_2N}, take any a and a net point b within α_0: $\Big|A_N(a) - \frac{1}{2}\|a\|^2\Big| \le 4Rd\alpha_0 + \varepsilon/2 + R\alpha_0 = R(4d+1)\alpha_0 + \varepsilon/2 \le \varepsilon$. So the bound holds with $C = 2M+1$ and $c = \min\{c_1, c_2\}$. ☐

7.6 A crude operator-norm bound

Lemma 7.9. $\mathbb{P}(\|W\|_{2\to 2} > 8) \le 2\exp(-(8 - \log 81)N)$.

Proof. If $\mathcal{M}$ is an ϵ-net of S^{N-1} with $\epsilon < \frac{1}{2}$, then $\|A\|_{2\to 2} \le (1-2\epsilon)^{-1} \max_{u,v\in\mathcal{M}} |u^T A v|$: for unit x, y and net points u, v within ϵ, $|y^T A x| \le |v^T A u| + |(y-v)^T A x| + |v^T A(x-u)| \le \max |p^T A q| + 2\epsilon \|A\|_{2\to 2}$, and take the supremum. A $\frac{1}{4}$-net has at most 9^N points by the volume argument. For fixed unit u, v, $u^T W v = \sum_{ij} u_i W_{ij} v_j$ is centred Gaussian of variance $\frac{1}{N}\sum_i u_i^2 \sum_j v_j^2 = 1/N$, so $\mathbb{P}(|u^T W v| > 4) \le 2e^{-8N}$ by (47); a union bound over at most 81^N pairs gives $\mathbb{P}(\|W\|_{2\to 2} > 8) \le 81^N \cdot 2e^{-8N}$, and $8 - \log 81 > 0$. $\square$

7.7 Proof of the deterministic theorem

Proof of Theorem 7.1. For $T = 0$ the bad event is empty. Let $T \ge 1$ and first $N \ge T+1$. On the event $\mathcal{O} = \{\|W\|_{2\to 2} \le 8\}$, whose complement has probability at most $C_{\text{op}} e^{-c_{\text{op}} N}$ by Lemma 7.9, $\|x_{t+1}\| = \|\phi(W x_t)\| \le \|W x_t\| \le 8\|x_t\|$, so $\|x_t\| \le 8^t \le 8^T =: R$ for all $t \le T$. Build the $d = T+1$ probes and innovations of the previous subsection; on $\mathcal{O}$ every coefficient vector satisfies $\|a_t\| = \|x_t\| \le R$. Put $\eta = \delta 2^{-(T+2)}$ and apply Lemma 7.8 with $d = T+1$, radius $R = 8^T$ and tolerance η: outside an event $\mathcal{E}^c$ of probability at most $C_{\text{LLN}} e^{-c_{\text{LLN}} N}$, the inequality $\left|A_N(a) - \frac{1}{2}\|a\|^2\right| \le \eta$ holds for every a in the ball at once. The coefficient vector a_t is random and depends on the Gaussians, and this is exactly why uniformity is needed: on $\mathcal{O} \cap \mathcal{E}$ the realized a_t lies in the ball, so it may be substituted, and (50) gives

$$\left|\|x_{t+1}\|^2 - \tfrac{1}{2}\|x_t\|^2\right| \le \eta, \qquad 0 \le t \le T-1.$$

Write $q_t = \|x_t\|^2$ and $e_t = |q_t - 2^{-t}|$; then $e_0 = 0$ and $e_{t+1} \le |q_{t+1} - \frac{1}{2}q_t| + \frac{1}{2}|q_t - 2^{-t}| \le \eta + \frac{1}{2}e_t$, so $e_t \le \eta(1 + \frac{1}{2} + \dots) \le 2\eta$, and $|2^t q_t - 1| = 2^t e_t \le 2^{T+1}\eta = \delta/2 < \delta$ for every $t \le T$. The bad event is therefore contained in $\mathcal{O}^c \cup \mathcal{E}^c$, of probability at most Ce^{-cN} with constants depending on T, δ only. For the finitely many $N < T+1$ the probability is at most 1, and enlarging C makes $Ce^{-cN} \ge 1$ there. $\square$

7.8 A random start, and a small exceptional set of inputs

Proof of Corollary 7.2. Condition on $X_0 = x$. Independence leaves the law of W unchanged, so (46) holds with the same constants for every x, and taking the expectation over X_0 gives the bound. $\square$

Proof of Theorem 7.3. Let $B(x_0, W)$ be the indicator that x_0 is bad for W. It is measurable: $(x_0, W) \mapsto x_t$ is continuous for each t, so $\max_t \left|2^t \|x_t\|^2 - 1\right|$ is continuous and its superlevel set is Borel. For deterministic μ_N put $M(W) = \int B(x_0, W)\, \mu_N(dx_0) = \mu_N(\mathcal{B}(W)) \in [0,1]$. By Tonelli and the uniform bound (46),

$$\mathbb{E}_W M(W) = \int \mathbb{P}_W\big(B(x_0, W) = 1\big)\, \mu_N(dx_0) \le Ce^{-cN},$$

and Markov's inequality with threshold $e^{-cN/2}$ gives $\mathbb{P}_W(M(W) > e^{-cN/2}) \le e^{cN/2} \cdot Ce^{-cN} = Ce^{-cN/2}$. For a random μ_N independent of W, condition on its realized value: the law of W is unchanged and the deterministic case applies with the same constants. $\square$

7.9 With a bias vector

Let $x_{t+1} = \phi(W x_t + b)$, with one bias vector reused at every step, independent of W, and $b_i \overset{\text{ind}}{\sim} \mathcal{N}(0, \sigma^2/N)$. The reference profile is

$$\bar{q}_0 = 1, \qquad \bar{q}_{t+1} = \tfrac{1}{2}(\bar{q}_t + \sigma^2), \qquad \text{so} \qquad \bar{q}_t = \sigma^2 + (1-\sigma^2)2^{-t},$$

whose fixed point is σ^2. The result below controls a fixed horizon; it makes no assertion about the infinite-time trajectory.

Theorem 7.10 (finite horizon, with a bias). *Fix an integer $T \geq 0$, $\delta > 0$ and $\sigma \geq 0$. There are $C, c > 0$ depending only on T, δ, σ such that for every $N \geq 1$ and every deterministic $x_0 \in S^{N-1}$,*

$$\mathbb{P}_{W,b}\Big(\max_{0\leq t\leq T}\big|\|x_t\|^2 - \bar{q}_t\big| > \delta\Big) \leq Ce^{-cN},$$

The analogues of Corollary 7.2 and Theorem 7.3 hold for a random start or input measure independent of the pair (W, b), with bad inputs defined by the displayed event.

Proof. The case $\sigma = 0$ follows from Theorem 7.1, and $T = 0$ is immediate. Suppose $\sigma > 0$, $T \geq 1$ and $N \geq T + 1$. Apply the innovation argument conditionally on b, taking $\sigma(b)$ as the initial information. Since W is independent of b, the conditional law of the $d = T + 1$ padded innovations is the same product Gaussian law for every b. Consequently those innovations are jointly independent of b after removing the conditioning.

Put $\beta_i = \sqrt{N}\, b_i$, $\Gamma_i' = (\Gamma_i, \beta_i/\sigma)$ and $a_t' = (a_t, \sigma)$. The vectors Γ_i' are therefore independent $\mathcal{N}(0, I_{d+1})$, and the driving identity becomes

$$\|x_{t+1}\|^2 = \frac{1}{N}\sum_{i=1}^{N}(a_t' \cdot \Gamma_i')_+^2, \qquad \|a_t'\|^2 = \|x_t\|^2 + \sigma^2.$$

On $\{\|W\|_{2\to 2} \leq 8,\ \|b\|^2 \leq 2\sigma^2\}$, the recursion $\|x_{t+1}\| \leq 8\,\|x_t\| + 2\sigma$ gives $\|a_t'\| \leq R' := 8^T(1+3\sigma)$ for $t \leq T$. The complement has probability at most Ce^{-cN} by Lemmas 7.5 and 7.9. Apply Lemma 7.8 in dimension $d + 1$ with radius R' and tolerance $\eta = \delta/4$. Its uniformity permits the random coefficient a_t' to depend on all the Gaussian vectors. Outside another event of probability at most Ce^{-cN},

$$\Big|\|x_{t+1}\|^2 - \tfrac{1}{2}(\|x_t\|^2 + \sigma^2)\Big| \leq \eta \quad (0 \leq t < T).$$

Hence $e_t := |\,\|x_t\|^2 - \bar{q}_t|$ satisfies $e_0 = 0$ and $e_{t+1} \leq \eta + e_t/2$, so $e_t \leq 2\eta < \delta$. Enlarge C for the finitely many $N < T + 1$. Finally, integrate the deterministic-input bound over a start or measure independent of (W, b) and apply Tonelli and Markov exactly as before, now over the joint randomness (W, b). □

7.10 What the theorem says and does not say

The factor $\frac{1}{2}$ is $\mathbb{E}(\xi_+)^2 = \frac{1}{2}\mathbb{E}\xi^2$ for a symmetric ξ: ReLU keeps half of the expected squared magnitude, and the theorem says that this average concentrates along the adaptively chosen trajectory. The horizon is fixed because the trajectory spans a subspace of dimension at most $T + 1$, the uniform law of large numbers is applied in that dimension, and the size of the net is then a constant independent of N; a horizon growing with N would make the net and the constants grow, and the present argument would need more. The deterministic theorem is uniform in the choice of one starting vector; it does not say that one matrix works for every point of the sphere at once, and Theorem 7.3 is the weaker simultaneous statement it does give: for one typical matrix, all but an exponentially small set of inputs are good. The start, or the measure, may be conditioned on only because it is independent of W; a start chosen after seeing the matrix could sit on an exceptional direction.

8 How the two results fit together

8.1 The same object, two questions

Both results are about a Gaussian matrix with independent $\mathcal{N}(0, \alpha/n)$ entries followed by ReLU, and both live at the scale where a layer keeps a squared norm of order one. Both rest on the same elementary fact, $\mathbb{E}(\xi_+)^2 = \frac{1}{2}\mathbb{E}\xi^2$ for a symmetric ξ: ReLU discards half of the expected squared magnitude, so a layer multiplies a typical squared norm by $\alpha/2$. In section 7 this is the number $\frac{1}{2}$ at $\alpha = 1$; in section 6 it is the reason the critical variance is 2. Both must handle dependence, but at different points. For a fixed forward input, the next independent layer is independent of the current activation. In the global problem the worst input and its masks are selected after all weights are seen. In the tied problem even a fixed start produces later states that depend on the reused matrix. Each proof treats the dependence relevant to its own quantifiers. Both replace an infinite supremum by a finite net and pay a fixed exponential price for it, both produce failure probabilities exponentially small in the width, and both fix the number of layers first and let the width grow, with the horizon T or the depth L entering the constants.

The questions are different, and the difference decides the method.

	global stability (Theorems 1.1 and 1.4)	finite horizon (Theorems 7.1 and 7.3)
matrices	L independent, one per layer	one matrix, applied T times
variance	α/n, any α	$1/N$, that is $\alpha = 1$
quantified over	every input at once (a supremum)	one input; then all but a small set
quantity controlled	the Lipschitz constant, a derivative	the norm of the trajectory
conclusion	$K_{n,L}(\alpha) \le 1$ w.h.p. for $\alpha < 2$, $L \ge L_0(\alpha)$	$\|x_t\|^2 \approx 2^{-t}$ for $t \le T$
the obstacle	the worst input's masks depend on the weights	x_t depends on W
the treatment	never condition: average the realizability event over the sign gauges	condition exactly: Gram–Schmidt probes give independent innovations
the counting	orthants met by an n-dimensional subspace of $\mathbb{R}^{nL}$, $\le 2e^{nLH(1/L)}$	a net of a ball in $\mathbb{R}^{T+1}$, size independent of N
the moment	one exact radial moment per layer, order $2\theta n$	a Bernstein bound for each net point
failure probability	$2e^{-c_L(\alpha)n}$ beyond $L_0(\alpha)$	Ce^{-cN} for every N

8.2 Why each method fits its question

The finite-horizon proof conditions on what the matrix has revealed, and it may do so because the revealed information is finite-dimensional: the trajectory queries W in at most $T + 1$ directions, Gram–Schmidt makes them orthogonal, and the innovation lemma says that orthogonal queries chosen from the past return independent Gaussian answers. The dimension of the queried subspace is the horizon, not the width, and the uniform law of large numbers is applied in that small dimension. The argument cannot be run over all inputs at once, because the set of all inputs queries W in every direction, and no fixed-length transcript describes it. This is the reason the deterministic theorem is about one input, and why the bound on the measure of exceptional inputs comes through Fubini, from a typical matrix's point of view, rather than through a supremum.

The global theorem asks about the worst input, so it cannot condition on a trajectory; there is no trajectory, there are 2^{nL} candidate activation patterns and the adversary picks among them after seeing the weights. The gauge argument refuses to condition at all. It fixes a pattern on paper, uses a symmetry of the law of the weights to average the realizability event over all 2^{nL} orthants, and turns the average into a deterministic count. The price is the count, $e^{nLH(1/L)}$ against the 2^{nL} patterns, and the argument works only when the radial contraction, one exact moment per independent layer, outweighs it, which is what the budget B_L measures and why a depth $L_0(\alpha)$ appears. The symmetry needs independent layers: the gauge inserts a sign matrix on the left of one layer and cancels it on the right of the next, and with a single tied matrix the available symmetry is conjugation $W \mapsto \Sigma W \Sigma$ by one sign matrix, whose orbit does not move a target orthant through all orthants. So the two methods are not interchangeable. One is a conditioning argument that works along a low-dimensional trajectory for one input; the other is a symmetry argument that works over all inputs for independent layers. What is quantified over decides which one applies.

8.3 Where the second method sharpens the first theorem

The expansive side of the global theorem, Theorem 1.3, is a one-input statement: at $\alpha > 2$ a fixed input's forward norm grows, so the Lipschitz constant exceeds one. It was proved with Chebyshev's inequality, which gives a $1/n$ rate. The finite-horizon proof handles the same one-input forward pass with exponential concentration, and its ingredient, the moment-generating function of a masked Gaussian square, is already in Lemma 5.5. Using it in the global setting gives an exponential rate.

Corollary 8.1 (the expansive side, exponentially). *For $\alpha > 2$ let $c = 1/\alpha < \frac{1}{2}$ and*

$$I(c) = \sup_{s>0}\Big[-sc - \log \frac{1 + (1+2s)^{-1/2}}{2}\Big] > 0.$$

Then for every $n \geq 1$ and $L \geq 1$,

$$\mathbb{P}\big(K_{n,L}(\alpha) \leq 1\big) \leq L\, e^{-nI(1/\alpha)}.$$

Proof. As in the proof of Theorem 1.3, $\mathbb{P}(K \leq 1) \leq L\,\mathbb{P}(S_n \leq n/\alpha)$ with $S_n = \sum_i B_i Z_i^2$. For $s > 0$, Markov's inequality on e^{-sS_n} and Lemma 5.5 with $t = -s$ give

$$\mathbb{P}(S_n \leq nc) = \mathbb{P}\big(e^{-sS_n} \geq e^{-snc}\big) \leq e^{snc}\Big[\frac{1 + (1+2s)^{-1/2}}{2}\Big]^n,$$

and optimizing over s gives $e^{-nI(c)}$. Positivity: $\log \frac{1+(1+2s)^{-1/2}}{2} = -\frac{s}{2} + O(s^2)$ as $s \downarrow 0$, so the bracket is $s(\frac{1}{2} - c) + O(s^2) > 0$ for small s when $c < \frac{1}{2}$. □

The rate $I(1/\alpha)$ is a number: section 9 lists it for several α and compares the two bounds. In the tied-weight dynamics the corresponding statement is Theorem 7.1 itself, and its constants come from the same Bernstein and Chernoff estimates; the only difference is the innovation step that makes the one-input pass legitimate when the matrix is reused.

8.4 What neither result says

The global theorem needs independent layers; the finite-horizon theorem needs one input and a fixed horizon. Whether the tied dynamics $x \mapsto \phi(Wx)$ is non-expansive as a map, that is, whether a statement like Theorem 1.4 holds for the global Lipschitz constant of $\phi(W\cdot)^{\circ L}$, is not settled by either argument: the gauge orbit of one matrix is too small, and a supremum over inputs is not a finite transcript. That question is open here, and it is the natural meeting point of the two methods.

9 Numerics

The calculations in this section are released with their generators, row parameters and outputs. Tables 1–7 are typeset directly from the JSON by `export_tables.py`: the exact quantities by evaluating the formulas, the Monte Carlo quantities by simulating the network with a fixed seed and comparing with the formulas they are supposed to obey. Where a quantity was sampled rather than enumerated the table says so, since a sampled supremum is a lower bound.

The finite transfer formulas of Theorem 10.4 also have a standard-library rational evaluator, `exact_region_transfer.py`. A separate calculation in `check_region_transfer.py` checks 3,072 actual selected-row ranks at $(n, L) = (2, 3), (3, 2)$, 1,020 scalar sign-orbit networks through depth eight, and both width-two formulas at depths one through 25. It rejects deliberate errors in selected/retained overlap, the final-mask factor, and the treatment of zero central rows versus empty affine constraints. The released JSON retains the integer matrices and source hashes. These finite author controls do not replace the almost-sure rank proof.

9.1 The orthant count

Table 1 draws a random subspace U of dimension d in $\mathbb{R}^M$ and counts the open orthants it meets. In the plane the routine enumerates boundary arcs using floating-point arithmetic; for $d \geq 3$ the count is a lower bound from sampled directions. The planar samples attain the bound of Lemma 4.3, in agreement with the general-position remark.

M	d	2^M	$2\sum_{j<d}\binom{M-1}{j}$	observed $r(U)$, mean (max)	method
6	2	64	12	12.0 (12)	arcs
8	2	256	16	16.0 (16)	arcs
10	2	1024	20	20.0 (20)	arcs
20	2	1048576	40	40.0 (40)	arcs
6	3	64	32	31.9 (32)	sampled
9	3	512	74	73.4 (74)	sampled
12	3	4096	134	131.8 (134)	sampled
8	4	256	128	120.8 (127)	sampled

Table 1: Orthants met by a random d-dimensional subspace of $\mathbb{R}^M$ against the bound of Lemma 4.3; 30 subspaces per row for $d = 2$, 12 for $d \geq 3$.

For the network itself, (20) with $F = 1$, summed over all masks, bounds the *expected* number of nonempty strict regions by $C_{n,L}$. This is not a deterministic bound on the number of regions of each network. Table 2 counts only masks with every preactivation nonzero at every layer. The separate zero column counts full-dimensional ordinary cells with at least one identically zero preactivation; boundary-only patterns enter neither count. At width two all cell boundaries are enumerated recursively. For $n = 3, 4$ every formal mask is tested by the homogeneous strict feasibility problem $As \geq \mathbf{1}$, after normalizing nonzero signed rows; zero rows exclude strict feasibility. Calculations use floating point, and returned LP witnesses are checked directly.

There is a useful exact check at $n = L = 2$. In coordinates $y = W_1 s$, the negative quadrant dies before layer two, each mixed quadrant contributes one strict cell, and the positive quadrant contributes one plus the number of rows of W_2 whose entries have opposite signs. Thus the strict count has law $3 + \mathrm{Binomial}(2, 1/2)$, with support $\{3, 4, 5\}$ and mean four. With $W_1 = I_2$ and $W_2 = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, boundary enumeration and a separate all-mask LP both give five strict cells and six ordinary cells. This control prevents the dead quadrant from entering the strict count. More generally, Proposition 10.8 gives the exact means $4, 4, 3.25, 2.5, 1.890625, 1.421875$ for the six width-two depths in the table. The displayed sample means fluctuate around those population values; the row at $L = 4$ is about 2.6 estimated standard errors above its mean, which alone does not refute the identity or diagnose a faulty standard error.

n	L	exact $\mathbb{E}R_{n,L}$	$C_{n,L}$	strict, sample mean (range)	zero, mean	method
2	1	4	4	4.00 (4–4)	0.00	arcs
2	2	4	8	3.93 (3–5)	1.00	arcs
2	3	13/4	12	3.57 (0–7)	1.72	arcs
2	4	5/2	16	3.17 (0–8)	2.47	arcs
2	5	121/64	20	1.68 (0–6)	3.78	arcs
2	6	91/64	24	1.77 (0–8)	3.97	arcs
3	2	29/2	32	15.16 (10–19)	1.00	LP
3	3	17	74	17.52 (6–35)	3.04	LP
4	2	52	128	52.32 (22–68)	1.00	LP

Table 2: Strict regions and excluded forced-zero cells (60 draws per row at $n = 2$, 25 otherwise). The exact mean is from Theorem 10.4; $C_{n,L}$ is its upper bound. Arcs enumerate planar cells; LP exhausts all 2^{nL} formal masks. Row seeds, weights, accepted masks and strict witnesses are released.

At fixed L the per-width exponent of $C_{n,L}$ converges to $LH(1/L)$ (Lemma 10.2). At $n = 40$: for $L = 3$, $\frac{1}{n}\log C_{n,L} = 1.851$ against $LH(1/L) = 1.910$ and $L\log 2 = 2.079$; for $L = 10$, 3.146 against 3.251 and 6.931; for $L = 100$, 5.434 against 5.600 and 69.31. The entropy bound of Lemma 4.4 is within a few percent of the exact count per unit width already at width 40, and the saving against the 2^{nL} formal masks is the whole difference between $\log L + 1$ and $L \log 2$.

9.2 The realizable-mask identity

Table 3 is the paired test of Proposition 4.6 at width $n = 2$ described in section 4: for a fixed mask D, the same weight draws are used on both sides and the paired difference is compared with its standard error. Two functionals are used, $F \equiv 1$ and $F = \min(\|J_D\|_{2\to2}, 1)$, both bounded. For the alternating masks in the table, all formal rows after the first layer are multiples of the first row of W_1. Almost surely, W_1 is invertible and these later rows are nonzero, so the two distinct first-layer boundary lines give exactly $r(\operatorname{im} A_D) = 4$. The right-hand observation therefore equals $4^{1-L}F$ on each draw.

The complete fixed-seed replay certifies this strict geometry and target membership for the real values represented by the sampled binary64 weights, using outward-rounded intervals with exact rational fallback. It also bounds the absolute error of every clipped norm by 10^{-12} using full matrix products and the rank-one identity $\|J_D\|^2_{2\to2} = \|J_D\|^2_{\mathrm{F}}$, including the final mask. The interval checks assume correctly rounded IEEE binary64 basic operations. The earlier rounded-angle routine disagreed on 210 of the 1,020,000 draws, including two target-membership decisions; the table uses the corrected results. The paired standard errors estimate sampling variation, separately from these arithmetic error bounds, and do not certify an ideal Gaussian sampler. The released reproduction note gives the two checks and their scope.

Two auxiliary checks in `s7_gauge.py` used the first 50 draws of each configuration. In each of those spot checks, summing the indicator $\mathbf{1}_{\{S_\Sigma U\cap O_D\neq\varnothing\}}$ over the whole gauge group returned $r(U)$, and $\|J_D\|_{2\to2}$ was unchanged by a random gauge to machine precision. These checks did not cover every Monte Carlo draw.

The all-active formula (55) gives $\mathbb{P}(R_D \neq \varnothing) = \frac{1}{2}$ at $(n, L) = (2, 2)$ and $(3, 2)$, 0.1445 at $(3, 3)$, 0.0898 at $(5, 3)$, 0.0307 at $(10, 3)$, 0.00432 at $(20, 3)$ and 3.0×10^{-18} at $(10, 10)$: the chance that a deep random network has an input at which every unit fires at every layer is exactly computable and small.

L	F	draws	$\mathbb{E}[F\mathbf{1}_{\{R_D\neq\varnothing\}}]$	$\mathbb{E}[F\,r(\operatorname{im} A_D)/2^{nL}]$	ratio	paired z
2	1	200000	0.24877	0.25000	0.995	−1.27
3	1	200000	0.06282	0.06250	1.005	+0.59
4	1	150000	0.01589	0.01562	1.017	+0.83
5	1	120000	0.00390	0.00391	0.998	−0.03
3	$\min(\lVert J_D\rVert_{2\to 2}, 1)$	200000	0.02307	0.02296	1.005	+0.42
4	$\min(\lVert J_D\rVert_{2\to 2}, 1)$	150000	0.00492	0.00493	0.999	−0.03

Table 3: Paired numerical checks of (20) at width 2, $D_\ell = \operatorname{diag}(1, 0)$ for odd ℓ and $D_\ell = I_2$ for even ℓ in every row; seed 8081 in `s7b_gauge_certified.py`. All six rows use the complete corrected replay described above.

9.3 The moment formulas

Table 4 checks the exact radial factor (29) and the mixture identity (31) by brute force.

identity	parameters	Monte Carlo	exact	ratio
(29)	$n = 4$, $L = 3$, $p = 1$, $\alpha = 1.3$, 1101,1001,1110	0.6247	0.6179	1.011
(29)	$n = 4$, $L = 3$, $p = 2$, $\alpha = 1.3$, 1101,1001,1110	2.063	2.121	0.973
(29)	$n = 6$, $L = 2$, $p = 1.5$, $\alpha = 1.3$, 101101,011011	0.9022	0.8987	1.004
(31)	$n = 6$, $p = 1$, $\alpha = 1.3$	41.65	41.6	1.001
(31)	$n = 6$, $p = 2.5$, $\alpha = 1.3$	61.99	61.93	1.001
(31)	$n = 12$, $p = 1.7$, $\alpha = 2$	5087	5079	1.002

Table 4: The fixed-vector radial moments and the scalar mixture identity, 2×10^5 draws for each radial row and 4×10^5 for each mixture row; the masks are the diagonals of $D_1, \ldots, D_L$; for (31) both sides are multiplied by 2^n.

The master inequality (32) is far from tight at small width, and the table says by how much: at $(n, L, \alpha, p) = (2, 2, 1, 1)$ the left side $\mathbb{E}K^{2p}$ is 1.517 against a bound of 200; at $(2, 3, 1, 1.5)$, 1.948 against 557; at $(2, 2, 0.6, 2)$, 0.882 against 318. The slack is the product of the net factor $5^n 4^p$ and of $C_{n,L}$ counting every orthant a subspace could meet rather than the ones it does. The inequality is designed for the exponential scale, where these factors cost a fixed rate per unit width, and it is there that it is used.

A finite-candidate greedy search produced $\frac{1}{2}$-separated sets of sizes 2, 9, 34, 102, 279 and 676 in dimensions $n = 1, \ldots, 6$, respectively. Maximality within a candidate pool does not establish coverage of the whole sphere. These empirical packing sizes provide no certified improvement to the 5^n covering bound in Lemma 5.2; the proof continues to use that deterministic bound.

9.4 The finite-width bound

Table 5 lists the threshold α_L^* of (43), the supremum of the α values for which the exact budget B_L is negative for some θ, together with the asymptotic formula of Theorem 1.4(iv) evaluated at the same L. The asymptotic expression alone does not certify a particular finite depth: at $L = 100$ it reads 1.01 with the $o(1)$ dropped, or 0.81 with the constant $A = 1 + \log 5 + 1$ kept, while the optimized finite budget delivers 0.954. At $L = 10^5$ the three agree to a few parts in a thousand.

The other reading of the same budget is the depth $L_0(\alpha)$ at which it first turns negative: 119 for $\alpha = 1$, 620 for 1.4, 3864 for 1.7, 50 026 for 1.9, 231 065 for 1.95 and 7 314 489 for 1.99. These are sufficient depths from the optimized scalar bound, computed by (38) and (39); they are not necessary depths for the networks. The certified lower bound α_L^* approaches two slowly,

L	α^*_L (exact budget)	$2e^{-\sqrt{10\log L/L}}$	$2e^{-\sqrt{10(\log L+A)/L}}$, $A = 2 + \log 5$
10	0.3670	0.4386	0.1758
30	0.6298	0.6896	0.4336
100	0.9537	1.0146	0.8080
300	1.2369	1.2932	1.1457
1000	1.4940	1.5378	1.4461
3000	1.6679	1.6986	1.6428
10000	1.7980	1.8170	1.7859
100000	1.9268	1.9333	1.9237
1000000	1.9746	1.9766	1.9738

Table 5: The threshold the finite theorem delivers at depth L, against the large-L formula.

and these computations do not give a matching rate for the true threshold α_L.

Table 6 gives the width rate $c_L(\alpha)$ of Corollary 1.2 and, in parentheses, the smallest integer width at which the bound $2e^{-c_L(\alpha)n}$ is at most 0.01. At $\alpha = 1$, $L = 300$ the bound is 3.0×10^{-47} at $n = 10$ and 1.6×10^{-234} at $n = 50$; at $L = 1000$ it is 1.8×10^{-238} at $n = 10$. These are values of the proven upper bound, not measured failure probabilities.

$\alpha\backslash L$	100	300	1000	3000	10^4	10^5
0.5	27.4 (1)	97.3 (1)	345 (1)	1.05e+03 (1)	3.54e+03 (1)	3.55e+04 (1)
1.0	–	10.8 (1)	54.8 (1)	183 (1)	634 (1)	6.45e+03 (1)
1.5	–	–	–	17.3 (1)	81.5 (1)	919 (1)
1.8	–	–	–	–	–	102 (1)
1.9	–	–	–	–	–	12.7 (1)

Table 6: The width rate $c_L(\alpha)$, with the width at which $2e^{-c_L(\alpha)n} \leq 0.01$ in parentheses; a dash means the budget is not negative at that depth.

On the other side, Theorem 1.3 gives $\mathbb{P}(K_{n,L}(\alpha) \leq 1) \leq 5\alpha^2 L/(n(\alpha-2)^2)$: at $\alpha = 3$ and $L = 10$ this is 0.45 at $n = 1000$ and 0.045 at $n = 10^4$; at $\alpha = 4$, $L = 10$, it is 0.2 and 0.02. The Chebyshev bound decays like $1/n$; the Chernoff form in section 8 decays exponentially.

9.5 What small networks do

Table 7 reports the measured Lipschitz constant $K_{n,L}(1)$ at small width. By the scaling identity (44), $K_{n,L}(\alpha) \leq 1$ exactly when $\alpha \leq K_{n,L}(1)^{-2/L}$, and the table reports $\widetilde{K}^{-2/L}$, the inverse transform of the sample median $\widetilde{K}$. This is a descriptive finite-sample statistic, not the width-limit threshold α_L. At $n = 2$ the constant is computed by enumerating all planar cells; for $n \geq 3$ it is a sampled lower bound, so the threshold column is an upper bound. At depth one the operator norm of W_1 approaches 2 at $\alpha = 1$ (measured 1.995 at $n = 2000$), so the threshold approaches $\frac{1}{4}$.

Two features of table 7 are worth reading against the theorem. At width two, deeper networks die: with probability growing in L some layer maps everything to zero, $K = 0$, and the reported inverse-median statistic can exceed 2, which is a small-width effect outside the scope of the threshold theorem. The fixed-width death rate is quantified separately in Corollary 10.6. At width eight and depth three the median threshold is about 0.6, well below 2, in keeping with the depth L_0 the budget demands before the bound takes effect. The theorem takes the width limit first and the depth limit second; the table is a reminder of what it does not say.

n	L	median $K_{n,L}(1)$	10%–90%	$\widetilde{K}^{-2/L}$	method
2	1	1.277	0.69–1.85	0.61	arcs
2	2	0.893	0.19–1.97	1.12	arcs
2	3	0.362	0.00–1.55	1.97	arcs
2	4	0.076	0.00–0.87	3.63	arcs
3	2	1.247	0.71–2.18	0.80	sampled
3	3	1.031	0.35–2.04	0.98	sampled
5	3	1.597	0.98–2.48	0.73	sampled
8	3	2.023	1.53–2.80	0.63	sampled

Table 7: Small-network Lipschitz constants at $\alpha = 1$. There are 400 draws for the first three rows, then 300, 150, 120, 60 and 40. Higher-dimensional rows use 40000 sampled directions per network; seeds and all measured K values are released.

9.6 One matrix, many steps

Table 8 measures the statistic of Theorem 7.1, the worst relative deviation $\max_{t\le 8}\left|2^t\left\|x_t\right\|^2 - 1\right|$ over eight steps, for the tied dynamics of section 7 and, beside it, for the feedforward pass with a fresh matrix at every step (the object of section 1 at $\alpha = 1$); 200 draws per width. Both columns shrink with the width, as both theorems say. The tied matrix wanders more than fresh matrices at this horizon: at $N = 2000$ six percent of the tied trajectories still stray by more than one half, against half a percent for fresh matrices. The theorem's rate is exponential in N with constants that depend on T, and eight steps of one matrix are not eight independent draws.

	one matrix			fresh matrix per step		
N	median	90%	frac. $> \frac{1}{2}$	median	90%	frac. $> \frac{1}{2}$
20	0.983	12.92	0.935	0.917	2.663	0.965
50	0.861	4.734	0.865	0.749	1.903	0.835
100	0.719	2.556	0.685	0.512	1.159	0.545
500	0.348	0.789	0.270	0.274	0.547	0.125
2000	0.192	0.434	0.060	0.137	0.243	0.005

Table 8: The worst deviation $\max_{t\le 8}|2^t\|x_t\|^2 - 1|$ over eight steps, 200 draws per width, seed fixed.

Table 9 evaluates the rate $I(1/\alpha)$ of Corollary 8.1 by a grid search over s, and compares the two bounds on $\mathbb{P}(K_{n,L}(\alpha) \le 1)$ at $L = 10$, $n = 1000$. Near $\alpha = 2$ both are useless at this width; from $\alpha = 3$ on the exponential bound is smaller by five, fifteen and thirty orders of magnitude.

α	$I(1/\alpha)$	Chebyshev, $5\alpha^2 L/(n(\alpha-2)^2)$	Chernoff, $Le^{-nI(1/\alpha)}$
2.2	0.00087	6.05	4.2
2.5	0.00455	1.25	0.11
3.0	0.01395	0.45	8.7×10^{-6}
4.0	0.03642	0.20	1.5×10^{-15}
6.0	0.07847	0.11	8.4×10^{-34}

Table 9: The expansive side at $L = 10$, $n = 1000$: the Chebyshev bound of Theorem 1.3 against the exponential bound of Corollary 8.1.

10 Sharpness and exact region counts

The finite theorem uses the elementary function g_α of Lemma 5.6. This section shows that the same function is the exact limiting moment exponent, so the scalar step loses no exponential rate at large width. It then records some structural consequences of the gauge identity.

10.1 The exact scalar high-moment exponent

For $\theta > 0$ and $0 \le u \le 1$ define

$$\Phi_{\alpha,\theta}(u) = H(u) - \log 2 + \theta \log(2\alpha) + \Big(\frac{u}{2} + \theta\Big) \log\Big(\frac{u}{2} + \theta\Big) - \frac{u}{2} \log \frac{u}{2} - \theta, \tag{51}$$

continuous on $[0, 1]$ because $x \log x$ extends continuously to zero.

For a chi-square variable with $m \ge 1$ degrees of freedom and $r > 0$, the substitution $x = 2y$ in $\int_0^\infty x^r \frac{x^{m/2-1} e^{-x/2}}{2^{m/2}\Gamma(m/2)}\, dx$ gives

$$\mathbb{E}[(\chi^2_m)^r] = 2^r \frac{\Gamma(m/2 + r)}{\Gamma(m/2)}. \tag{52}$$

Conditioning on M_n in (31) with $r = \theta n$ (the term $m = 0$ contributes zero),

$$\mathbb{E}[(X_n^{(\alpha)})^{\theta n}] = 2^{-n} \sum_{m=1}^{n} \binom{n}{m} \Big(\frac{2\alpha}{n}\Big)^{\theta n} \frac{\Gamma(m/2 + \theta n)}{\Gamma(m/2)}. \tag{53}$$

Two forms of Stirling's formula are used: for integers $k \ge 1$, $\log k! = (k + \frac{1}{2}) \log k - k + O(1)$, and for real $x \ge \frac{1}{2}$, $\log \Gamma(x) = (x - \frac{1}{2}) \log x - x + O(1)$ uniformly in $x \ge \frac{1}{2}$ (the function $\log \Gamma(x) - (x - \frac{1}{2}) \log x + x$ is continuous on $[\frac{1}{2}, \infty)$ and tends to $\frac{1}{2} \log 2\pi$). With $u = m/n$ they give, uniformly for $1 \le m \le n$,

$$\log \binom{n}{m} = nH(u) + O(\log n),$$

$$\log \Gamma\Big(n\big(\tfrac{u}{2} + \theta\big)\Big) - \log \Gamma\Big(\frac{nu}{2}\Big) = \theta n \log n + n\Big[\big(\tfrac{u}{2} + \theta\big) \log\big(\tfrac{u}{2} + \theta\big) - \tfrac{u}{2} \log \tfrac{u}{2} - \theta\Big] + O_\theta(\log n).$$

The term $-\theta n \log n$ from $(2\alpha/n)^{\theta n}$ cancels the $+\theta n \log n$ of the Gamma ratio, and the m-th summand $a_{n,m}$ of (53) satisfies $\frac{1}{n} \log a_{n,m} = \Phi_{\alpha,\theta}(m/n) + O_{\alpha,\theta}(\log n/n)$ uniformly. A sum of n nonnegative terms lies between its largest term and n times it, so after taking $\frac{1}{n}\log$ the sum and the maximum differ by at most $\log n/n$; and the grids $\{1/n, \ldots, 1\}$ become dense, so the grid maximum of the continuous $\Phi_{\alpha,\theta}$ converges to its maximum.

Proposition 10.1 (exact exponent)**.** *For every $\alpha > 0$ and $\theta > 0$,*

$$\lim_{n\to\infty} \frac{1}{n} \log \mathbb{E}[(X_n^{(\alpha)})^{\theta n}] = \max_{0 \le u \le 1} \Phi_{\alpha,\theta}(u) = g_\alpha(\theta),$$

where g_α is the finite-width bound in (4). The unique maximizing fraction is $u_\theta = 1/(1 + z_\theta)$.

Proof. The limit and its variational formula follow from the exact Gamma sum and uniform estimates above. For $0 < u < 1$,

$$\partial_u \Phi = \log \frac{1-u}{u} + \frac{1}{2} \log \frac{u/2 + \theta}{u/2}, \qquad \partial_{uu} \Phi = -\frac{1}{u(1-u)} + \frac{1}{4(u/2 + \theta)} - \frac{1}{2u} < 0.$$

The first derivative tends to $+\infty$ at zero and $-\infty$ at one, so there is one interior maximizer. Substitute $u = 1/(1 + z)$. Its stationary equation reduces to $\theta = (1 - z)/(2z^2)$, giving $z = z_\theta$ and

$$\frac{u_\theta}{2} + \theta = \frac{1}{2z_\theta^2(1 + z_\theta)}.$$

Substituting these identities into (51) and collecting logarithms gives exactly (4). This identifies the limiting exponent with the function that already bounds every finite-width moment in Lemma 5.6. $\square$

In particular $u_\theta = \frac{1}{2} + \frac{1}{2}\theta + O(\theta^2)$ and (37) gives

$$g_{2e^{-\delta}}(\theta) = -\delta\theta + \frac{5}{2}\theta^2 + O(\theta^3). \tag{54}$$

The scalar exponent is exact, but the geometric count and the sphere net in the master inequality still introduce slack. The constant $\sqrt{10}$ remains a constant in a certified lower bound for the network threshold, not a matching asymptotic description of that threshold.

10.2 The Cover factor at fixed depth

Lemma 10.2. *For every fixed $L \geq 2$, $\lim_{n\to\infty} \frac{1}{n}\log C_{n,L} = LH(1/L)$.*

Proof. For $L > 2$ the ratio of consecutive terms $\binom{nL-1}{j+1}/\binom{nL-1}{j} = (nL-1-j)/(j+1)$ exceeds 1 for $0 \leq j \leq n-2$, because $nL-1-j \geq nL-n+1 > n-1 \geq j+1$; so the last term is the largest and $\binom{nL-1}{n-1} \leq C_{n,L}/2 \leq n\binom{nL-1}{n-1}$. Stirling's formula gives $\frac{1}{n}\log\binom{nL-1}{n-1} \to LH(1/L)$, and the factors 2 and n vanish after $\frac{1}{n}\log$. For $L = 2$, $C_{n,2} = 2^{2n-1}$ and $\frac{1}{n}\log C_{n,2} \to 2\log 2 = 2H(\frac{1}{2})$. $\square$

So the entropy bound of Lemma 4.4 is exact at the exponential scale, and the finite theorem loses nothing there at large width. At small width it does lose something; section 9 shows how much.

10.3 The all-active pattern, exactly

Take $D_1 = \cdots = D_L = I_n$. Then $H_\ell^D = Q_\ell := W_\ell \cdots W_1$ and A_D stacks $Q_1, \ldots, Q_L$. Every Gaussian square matrix is invertible almost surely, since its determinant is a nonzero polynomial in continuously distributed entries and the zero set of a nonzero polynomial has Lebesgue measure zero (Lemma 10.3). On that event $(W_1, \ldots, W_L) \mapsto (Q_1, \ldots, Q_L)$ is a smooth bijection with inverse $W_1 = Q_1$, $W_\ell = Q_\ell Q_{\ell-1}^{-1}$, so the tuple (Q_ℓ) has a joint density on $(GL_n)^L$. Choose any n of the nL rows of the stacked matrix; their determinant is a polynomial in the entries of the Q_ℓ, not identically zero (assign the chosen rows the standard basis vectors and complete each Q_ℓ to an invertible matrix), hence nonzero almost surely; finitely many choices, so all $n \times n$ minors are nonzero almost surely. That is central general position for the coordinate hyperplanes restricted to $\operatorname{im} A_D$, and the region count is then exact: $r(\operatorname{im} A_D) = C_{n,L}$ almost surely. Proposition 4.6 with $F = 1$ gives

$$\mathbb{P}(R_D \neq \varnothing) = \frac{C_{n,L}}{2^{nL}} = 2^{1-nL}\sum_{j=0}^{n-1}\binom{nL-1}{j} \qquad (D \text{ all-active}), \tag{55}$$

an exact formula for the probability that a random deep network has an input at which every unit is active at every layer. Its logarithm is $-[nL\log 2 - \log C_{n,L}] = -n[L\log 2 - LH(1/L)] + o(n)$ by Lemma 10.2: the accumulated cost per unit width is $L\log 2 - O(\log L)$, not a constant per layer. For $L = 1$ the cost is zero, as it must be: $C_{n,1} = 2^n$, and a full-rank W_1 reaches every orthant. Section 9 checks (55) against direct enumeration.

Lemma 10.3. *If $P : \mathbb{R}^d \to \mathbb{R}$ is a polynomial that is not identically zero, then $\{x : P(x) = 0\}$ has d-dimensional Lebesgue measure zero.*

Proof. Induction on d. For $d = 1$ a nonzero polynomial has finitely many roots. For $d > 1$ write $P = \sum_{k=0}^m a_k(x_1, \ldots, x_{d-1})x_d^k$ with some a_k not identically zero. By induction the set where all a_k vanish has measure zero; for every other value of the first $d-1$ coordinates the polynomial in x_d is nonzero and has finitely many roots, a one-dimensional null set. Fubini's theorem gives total measure zero. $\square$

10.4 An exact finite formula for the mean region count

The gauge identity gives more than an upper bound on the number of strict regions. Their mean can be computed from a triangular matrix whose size depends only on the width. The same matrix treats independent centred nondegenerate Gaussian biases by restoring one extra index.

Fix $n \geq 1$, put $q = 2^n$, and define the binomial tails

$$\mathcal{B}_d(t) = \sum_{j=t}^{d} \binom{d}{j}, \qquad 0 \leq t \leq d.$$

Define a matrix $M = M_n$, with rows and columns indexed by $0, \dots, n$, by

$$M_{r,t} = \begin{cases} 2^{r-t}\binom{n}{r-t}\mathcal{B}_{n-r+t}(t), & 0 \leq t \leq r \leq n, \\ 0, & r < t. \end{cases} \tag{56}$$

Let U be its principal submatrix on $1, \dots, n$, and let

$$c_t = \binom{n-1}{t-1} \quad (1 \leq t \leq n), \qquad d_t = \binom{n}{t} \quad (0 \leq t \leq n).$$

All entries are nonnegative integers.

Theorem 10.4 (exact mean strict-region counts)**.** *Consider the square independent Gaussian network of (1), with input domain $\mathbb{R}^n$, positive variance parameter α, and ReLU at every layer, including the output. For every $L \geq 1$, its zero-bias strict-region count satisfies*

$$\mathbb{E}R_{n,L} = 2q^{1-L}\mathbf{1}^T U^{L-1} c. \tag{57}$$

If instead all bias coordinates are independent centred Gaussian variables of positive variances, independent of every weight, then the affine strict-region count satisfies

$$\mathbb{E}R^{(b)}_{n,L} = q^{1-L}\mathbf{1}^T M^{L-1} d. \tag{58}$$

The bias variances may differ across coordinates and layers. These counts retain different activation patterns even when the final affine output maps agree.

The finite-dimensional rank calculation behind both identities is useful on its own.

Lemma 10.5 (rank of a formal stack)**.** *Fix intermediate masks $D_1, \dots, D_{L-1}$ and selected row sets $S_j \subseteq \{1, \dots, n\}$ from the formal preactivation matrices H_j^D. Write $s_j = |S_j|$ and identify each mask with its set of retained coordinates. Almost surely the rank of all selected stacked rows is r_1, where*

$$r_L = s_L, \qquad r_j = s_j + \min\{r_{j+1}, |D_j \setminus S_j|\}. \tag{59}$$

At any fixed depth this holds simultaneously for every formal mask and every row selection.

Proof. Work backwards in coefficient space. Set $V_L = \operatorname{span}\{e_i^T : i \in S_L\}$ and

$$V_j = \operatorname{span}\{e_i^T : i \in S_j\} + V_{j+1}W_{j+1}D_j.$$

Then V_{j+1} depends only on $W_{j+2}, \dots, W_L$ and the fixed masks and selections, so it is independent of W_{j+1}. Conditional on that future, choose an orthonormal row basis of V_{j+1}. Multiplication by W_{j+1} gives a Gaussian matrix with independent nondegenerate columns. Its restriction to the $|D_j \setminus S_j|$ coordinates outside S_j has rank $\min\{\dim V_{j+1}, |D_j \setminus S_j|\}$ almost surely. Adding the s_j selected coordinate vectors proves (59). The span of the original selected rows is V_1W_1; W_1 is invertible almost surely and preserves its rank. A finite union gives the simultaneous assertion. No activation or realizability event was conditioned on. □

Proof of Theorem 10.4. For a fixed intermediate-mask sequence, let r_D^0 be the number of open orthants met by its central preactivation stack. Whitney's characteristic-polynomial formula and Zaslavsky's region formula give

$$r_D^0 = \sum_{S_1,\dots,S_L} (-1)^{s_1+\cdots+s_L-r_1}. \tag{60}$$

See Stanley [7], Theorems 2.4 and 2.5(11). Indexed repetitions of a nonzero hyperplane cause no problem: the alternating sum over all nonempty subsets of its copies is -1. If a row is zero, the strict orthant count is zero, and the sum in (60) also vanishes by pairing selections that differ only in that row. Thus the identity includes every formal mask, without treating the whole ambient space as a proper hyperplane.

Summing (20) with functional 1 cancels the q choices for the final mask, which does not enter the stack:

$$\mathbb{E}R_{n,L} = q^{1-L} \sum_{D_1,\dots,D_{L-1}} \mathbb{E}r_D^0. \tag{61}$$

For a future rank t and a current row selection of size s, there are $\binom{n}{s}$ choices of selected coordinates, 2^s choices for the mask inside them, and $\binom{n-s}{j}$ choices for j retained coordinates outside them. The backward rank is $s+\min(t,j)$. Consequently (60) and (61) give

$$\mathbb{E}R_{n,L} = q^{1-L} w^T A^{L-1} v, \quad v_t = (-1)^t \binom{n}{t}, \quad w_r = (-1)^r,$$

where the signed transfer matrix on $0,\dots,n$ is

$$A_{r,t} = \sum_{s=0}^{n} \sum_{j=0}^{n-s} (-2)^s \binom{n}{s} \binom{n-s}{j} \mathbf{1}_{\{r=s+\min(t,j)\}}.$$

A change of polynomial basis removes the signs. Associate a vector with its coefficient polynomial in x, and write $f_t(x)$ for column t of A. Then $f_t(1)=0$, and for $1 \le t \le n$,

$$f_t(x) - f_{t-1}(x) = (x-1)x^{t-1} \sum_{s=0}^{n-t} (-2)^s \binom{n}{s} \mathcal{B}_{n-s}(t) x^s.$$

Indeed, $\min(t,j)$ differs from $\min(t-1,j)$ exactly when $j \ge t$. In the basis $e_t(x) = (x-1)x^{t-1}$ of the polynomials vanishing at 1, the matrix therefore has entries

$$(-2)^{r-t} \binom{n}{r-t} \mathcal{B}_{n-r+t}(t), \qquad 1 \le t \le r \le n.$$

Conjugation by $\operatorname{diag}((-1)^t)$ gives U. The initial polynomial $(1-x)^n$ has e_t-coefficient $(-1)^t c_t$, while $e_t(-1) = 2(-1)^t$. Evaluation at -1 proves (57).

For biases, the formal affine preactivations are $H_j^D s + a_j$, with $a_1 = b_1$ and $a_j = b_j + W_j D_{j-1} a_{j-1}$. Conditional on all weights and a fixed mask sequence, the map from $(b_1,\dots,b_L)$ to $(a_1,\dots,a_L)$ is block lower triangular with identity diagonal. Thus all formal offsets have a joint density. They need not be independent.

A selected family of affine hyperplanes has nonempty intersection almost surely exactly when its normal rows are linearly independent: independent rows define an onto linear map, while consistent offsets for dependent rows lie in a proper linear subspace. A zero normal row has a nonzero constant almost surely and contributes no hyperplane. The affine versions of Whitney's and Zaslavsky's formulas therefore count independent subsets of the normal rows, since every consistent selected subset has its size equal to its rank.

Suppose the future contains t independent selected rows. For $0 \le s \le n-t$, selecting s current rows preserves independence of all $t+s$ rows precisely when at least t mask coordinates outside the selection are retained, by Lemma 10.5. There are

$$\binom{n}{s} 2^s \mathcal{B}_{n-s}(t) = M_{t+s,t}$$

choices; selecting more than $n-t$ current rows cannot preserve independence. At the final block the initial counts are $d_t = \binom{n}{t}$. The affine gauge identity from section 6.8, with the same cancelled final-mask factor q, now proves (58). $\square$

10.5 Depth asymptotics at fixed width

Keep n fixed and put $\beta_n = 1 - 2^{-n}$. Let $S_{n,L} = \{F_{n,L}^{(\alpha)} \not\equiv 0\}$ denote global survival. A strictly negative final preactivation can leave a nonempty strict region whose output is zero, so survival and strict-region counts are different objects.

Corollary 10.6 (fixed-width depth asymptotics)**.** *For the zero-bias network, there are finite positive constants κ_n, σ_n, independent of $\alpha > 0$, such that*

$$\mathbb{E}R_{n,L} \sim \kappa_n \beta_n^L, \qquad \mathbb{P}(S_{n,L}) \sim \sigma_n \beta_n^L. \tag{62}$$

Both normalized ratios are nondecreasing. The mean constant is explicit: set $m = 2^n - 1$, $u_1 = 1$, and

$$u_r = \frac{\sum_{t=1}^{r-1} U_{r,t} u_t}{m - \mathcal{B}_n(r)} \quad (2 \le r \le n), \qquad \kappa_n = \frac{2^{n+1}}{2^n - 1} \sum_{r=1}^{n} u_r. \tag{63}$$

For $n \ge 2$ the mean also has the remainder estimate

$$\frac{\mathbb{E}R_{n,L}}{\beta_n^L} = \kappa_n + O_n\left(\left[\frac{\mathcal{B}_n(2)}{2^n - 1}\right]^L\right).$$

Moreover $\kappa_n \ge n\beta_n^{-2}$ and $\beta_n^{-1} \le \sigma_n \le \kappa_n$. In particular,

$$\lim_{L\to\infty} \frac{\log \mathbb{E}R_{n,L}}{L} = \lim_{L\to\infty} \frac{\log \mathbb{P}(S_{n,L})}{L} = \log(1 - 2^{-n}). \tag{64}$$

Proof. The diagonal entries of U are the distinct positive tails $\mathcal{B}_n(1), \dots, \mathcal{B}_n(n)$. Its largest eigenvalue is $m = \mathcal{B}_n(1)$, with positive right eigenvector u given by (63). Every other right eigenvector has first coordinate zero, whereas $c_1 = 1$. Thus the coefficient of u in the spectral decomposition of c is 1. Apply (57). The other eigenvalues are at most $\mathcal{B}_n(2) < m$, proving the mean limit and its remainder. At $n = 1$ the matrix is $U = (1)$ and the same formula gives $\kappa_1 = 4$ directly.

For monotonicity, let N_L count strict regions whose final mask is nonzero. In the gauge identity the stack is independent of the choice of final mask, so $\mathbb{E}N_L = \beta_n \mathbb{E}R_{n,L}$. Conditional on the first L layers, each such region has postactivation map $x_L(s) = As$, where $A = D_L H_L^D \neq 0$ because at least one retained coordinate is strictly positive. A fresh Gaussian row w satisfies $wA \neq 0$ almost surely. Removing the next layer's proper hyperplanes therefore leaves at least one nonempty open strict subregion in each of the finitely many regions counted by N_L. Different old activation patterns remain different after extension, so $R_{n,L+1} \ge N_L$ almost surely. Taking expectations proves monotonicity of the normalized means.

For survival, couple all depths by one infinite independent sequence and put $p_L = \mathbb{P}(S_{n,L})$. On $S_{n,L}$ choose the first input in a fixed enumeration of $\mathbb{Q}^n$ with nonzero output. It exists by continuity and is measurable with respect to the first L matrices. Conditional on that past, the next fresh layer preserves its output with probability exactly β_n, so $p_{L+1} \geq \beta_n p_L$. Positive homogeneity gives $F(0) = 0$, and Lemma 3.5 ensures that a nonzero network has a realized strict region. Hence $p_L \leq \mathbb{E}R_{n,L}$, and p_L/β_n^L is nondecreasing and bounded above by κ_n. Since $p_1 = 1$, its limit lies in $[\beta_n^{-1}, \kappa_n]$. The lower bound on κ_n follows from (65) below. Positive rescaling of the weights changes neither signs nor zero-output events, so the constants are independent of $\alpha > 0$. □

Some useful finite bounds remain particularly simple:

$$n\beta_n^{L-2} \;\leq\; \mathbb{E}R_{n,L} \;\leq\; C_{n,L}\beta_n^{L-1}, \qquad L \geq 2, \tag{65}$$

$$\beta_n^{L-1} \;\leq\; \mathbb{P}(S_{n,L}) \;\leq\; \min\{1, C_{n,L}\beta_n^L\}, \qquad L \geq 1. \tag{66}$$

For the mean lower bound, take a rank-one first mask and nonzero remaining intermediate masks. There are $n(q-1)^{L-2}$ such sequences. All later formal rows are nonzero multiples of a row of W_1 almost surely, so their central orthant count is q; (61) gives the result. For the upper bound, a zero intermediate mask kills the strict count, leaving $(q-1)^{L-1}$ sequences, each with count at most $C_{n,L}$. For the survival upper bound, use the invariant functional $\mathbf{1}_{\{J_D \neq 0\}}$ in Proposition 4.6; a zero mask at any layer, including the last, kills the Jacobian, leaving at most $(q-1)^L$ contributors. For its lower bound choose $s = W_1^{-1}e_1$ and propagate the first-layer output e_1 through the remaining independent layers.

At width one, $\mathbb{E}R_{1,L} = 2^{2-L}$ and $\mathbb{P}(S_{1,L}) = 2^{1-L}$, so $(\kappa_1, \sigma_1) = (4, 2)$. The next mean constants are $\kappa_2 = 8$ and $\kappa_3 = 240/7$. No exact higher-width survival constant, convergence rate for the survival ratio, or uniformity in a growing width is asserted.

Earlier work already establishes death at large depth. Rister and Rubin [10], Sections 3–4, give global and fixed-input survival bounds with respective bases $1 - 2^{-n^2}$ and β_n in the zero-bias square setting; their Section 5.3 also treats a finite-data survival ratio as width grows, under a variance assumption. Lu et al. [9], Theorem 3.5 and Appendix C, give two-exponential bounds and a one-ray state analysis for one-dimensional input with a final affine readout. After adjusting that depth convention, their bounds and the witness argument above also imply a finite positive survival-ratio limit in the scalar-input setting. A fixed finite set of M inputs needs only $p_L \leq M\beta_n^L$ and the same monotonicity. The finite transfer formula supplies the corresponding geometric control over the entire square input space. Neither qualitative eventual death nor ratio arguments in general are claimed as new.

10.6 The finite limiting mean with independent biases

Corollary 10.7. *Under the nondegenerate Gaussian bias assumptions of Theorem 10.4, set $z_0 = 1$ and*

$$z_r = \frac{\sum_{t=0}^{r-1} M_{r,t} z_t}{2^n - \mathcal{B}_n(r)} \quad (1 \leq r \leq n), \qquad \rho_n = \sum_{r=0}^{n} z_r.$$

Then ρ_n is a finite positive rational number and

$$\mathbb{E}R_{n,L}^{(b)} = \rho_n + O_n(\beta_n^L).$$

On one infinite independent-layer sequence the geometric open strict cells eventually stop changing almost surely. The number of cells in that eventual partition has expectation ρ_n.

Proof. The eigenvalues of M are the distinct tails $\mathcal{B}_n(0), \ldots, \mathcal{B}_n(n)$. The largest is $q = 2^n$, with right eigenvector z; since $d_0 = z_0 = 1$, its spectral coefficient is 1. All remaining eigenvalues are at most $q - 1$, so (58) gives the limit and remainder.

With nondegenerate independent biases, each new preactivation on an old open cell is almost surely either a nonconstant affine function, whose zero set is the intersection of the cell with a proper affine hyperplane, or a nonzero constant. Thus the strict partitions refine: their counts are nondecreasing, and any change of their geometric open cells increases the count. Monotone convergence and the finite mean limit imply a finite integer limiting count almost surely, so only finitely many refinements occur and its expectation is ρ_n. □

The first limits are $\rho_1 = 3$, $\rho_2 = 21$ and $\rho_3 = 279$. The finite expectation is stronger than qualitative stabilization alone. After the first layer, a layer with all weights and biases negative maps the whole nonnegative image to zero; it has probability 2^{-n^2-n} independently at each depth. Once this happens, later layers cannot recover input dependence. This elementary dropout mechanism also appears in the discussion of Wang [8], Section 4.1. The stabilized objects here are geometric activation cells. Their depth-indexed labels and the constant output may keep changing. The exact affine mean requires nondegenerate biases and must not be obtained by substituting zero variance.

10.7 The width-two formulas

Proposition 10.8. *For every $L \geq 1$ and $\alpha > 0$,*

$$\mathbb{E}R_{2,L} = 8\left[\left(\frac{3}{4}\right)^L - \left(\frac{1}{4}\right)^L\right], \tag{67}$$

$$\mathbb{E}R^{(b)}_{2,L} = 21 - 24\left(\frac{3}{4}\right)^L + 4\left(\frac{1}{4}\right)^L, \tag{68}$$

where the second identity uses the bias assumptions of Theorem 10.4.

Proof. At width two,

$$U = \begin{pmatrix} 3 & 0 \\ 4 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} 4 & 0 & 0 \\ 8 & 3 & 0 \\ 4 & 4 & 1 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad d = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

Evaluating the powers in (57) and (58) gives the two identities, including $L = 1$ when both means are 4. □

The first six zero-bias means are $4, 4, 3.25, 2.5, 1.890625, 1.421875$. Their decay concerns strict regions; ordinary cells remain when the network is identically zero. With nondegenerate biases, the strict partition instead refines and its expected count increases to 21. These are activation counts throughout, rather than counts of maximal regions of a single output formula.

10.8 Depth one

At $L = 1$ the all-active mask is realized with probability one, $J_D = W_1$ for it, and (14) gives $K_{n,1}(\alpha) = \|W_1\|_{2\to 2}$: every mask D_1 gives $\|D_1 W_1\|_{2\to 2} \leq \|W_1\|_{2\to 2}$. By the Bai–Yin theorem [15], for the centered Gaussian matrix of this paper, with independent $\mathcal{N}(0, \alpha/n)$ entries, $\|W_1\|_{2\to 2} \to 2\sqrt{\alpha}$ almost surely. Hence $\mathbb{P}(K_{n,1}(\alpha) \leq 1) \to 1$ for $\alpha < \frac{1}{4}$ and $\to 0$ for $\alpha > \frac{1}{4}$, that is,

$$\alpha_1 = \tfrac{1}{4}.$$

A depth-one network at $\alpha = 1$ is expansive with probability tending to one, although $1 < 2$; there is no contradiction with Theorem 1.4, whose part (i) only takes effect beyond $L_0(\alpha)$.

10.9 Why the global upper bound never conditions on an input

A tempting route would be to choose an input, condition on the activation masks it generates, and analyse the resulting Jacobian as if the masks were independent of the weights. The last step is false in general: the event that a mask is generated by an input carries information about the same weights that appear in the Jacobian. The gauge argument fixes a formal mask without choosing a witness input, multiplies the realizability indicator by a gauge-invariant Jacobian functional, and averages over a finite symmetry group exactly; only then does it sum over masks. The sphere net is used for tangent directions of a fixed linear map, not to search for regions, so a realizable cone of arbitrarily small angle is still counted.

10.10 The order of limits

The master inequality (32) and the finite theorem are statements about every n and L, and one may evaluate them in any joint regime. What is proved about the thresholds, however, fixes L first. If $L = L(n)$ grows rapidly, several effects change scale; for instance the probability that some layer dies at a fixed input is at most $L2^{-n}$, negligible for fixed L but not for every $L(n)$. No joint-limit statement should be read into Theorem 1.4 beyond what Theorem 1.1 gives when its budget is evaluated in that regime.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author made substantial use of generative AI tools, and describes their role here. The two theorems and their proofs are the author's, from two earlier manuscripts. The present text was rewritten and consolidated with Fable 5.1 (Anthropic, through the Claude Code agent, September 2026): the finite-width formulation of the first theorem and its computable threshold, the condensed account of the finite-horizon proof, the section comparing the two arguments and the corollary that sharpens the expansive side, the extension to biases, the numerical experiments and the code behind every table, and the drafting of the exposition. Subsequent Codex-assisted revisions corrected the strict-region enumeration, checked the affected numerical calculations, distinguished sufficient certificates from necessary conditions, added a comparison with related work, and repaired the bias hypotheses and proofs. They also derived the optimized scalar moment bound, exact finite matrix formulas for mean strict-region counts with and without independent Gaussian biases, and the fixed-width mean and survival asymptotics, with separate independent checks of the new mathematics. The released source and code record these changes and their verification scope. The author takes responsibility for the manuscript; computational checks do not replace mathematical review.

Data and code availability

The source archive accompanying this version contains the code behind every table, the fixed-seed simulation outputs, and the scripts that evaluate the exponent budget, the thresholds α_L^* and the rates $I(1/\alpha)$. The project repository is `https://github.com/yspennstate/relu-global-stability-numerics`.

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